

BFS Traversal

Step 1: Start at E

Legend

Complete

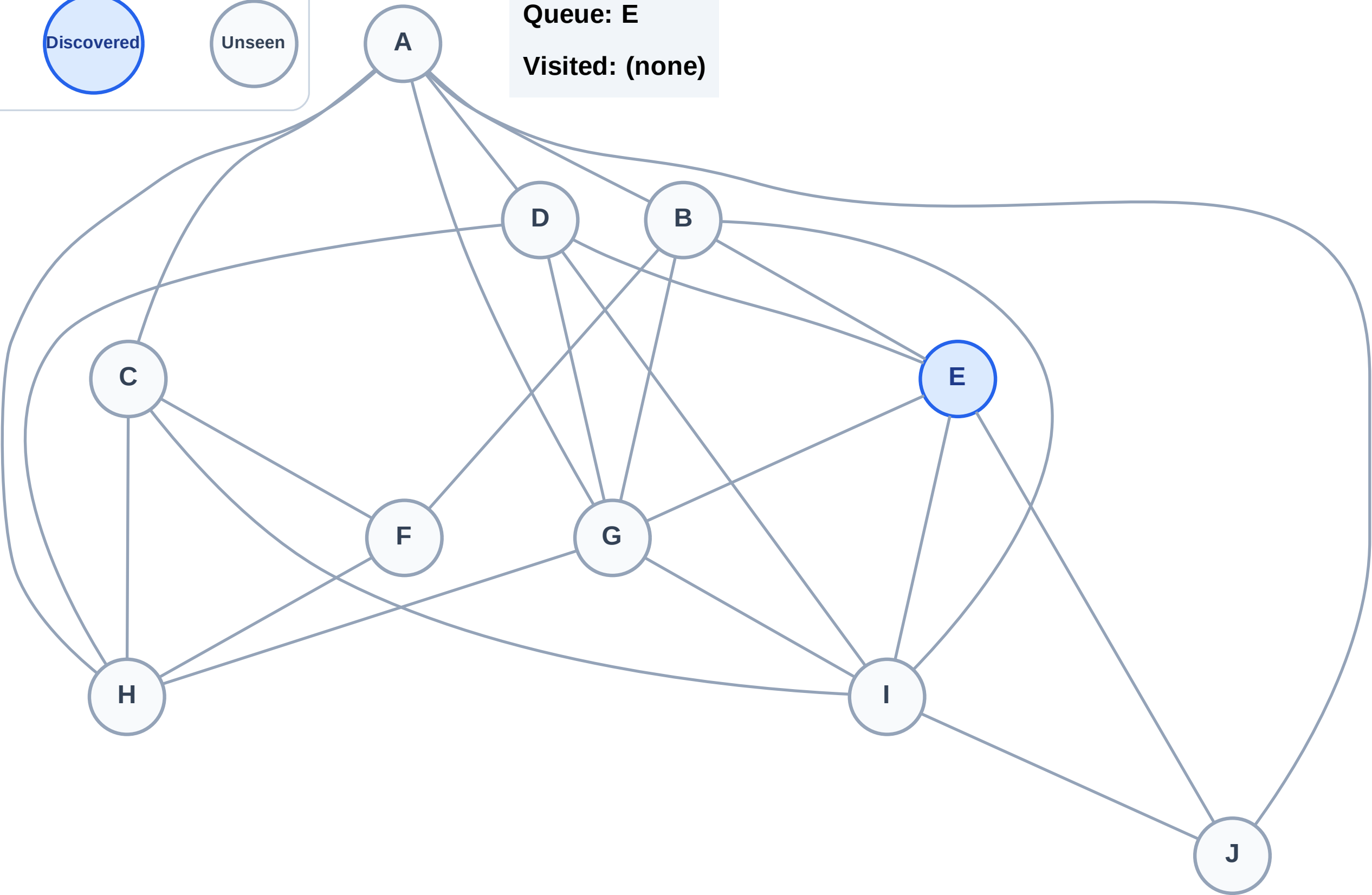
Processing

Discovered

Unseen

Queue: E

Visited: (none)



BFS Traversal

Step 2: Process node E

Legend

Complete

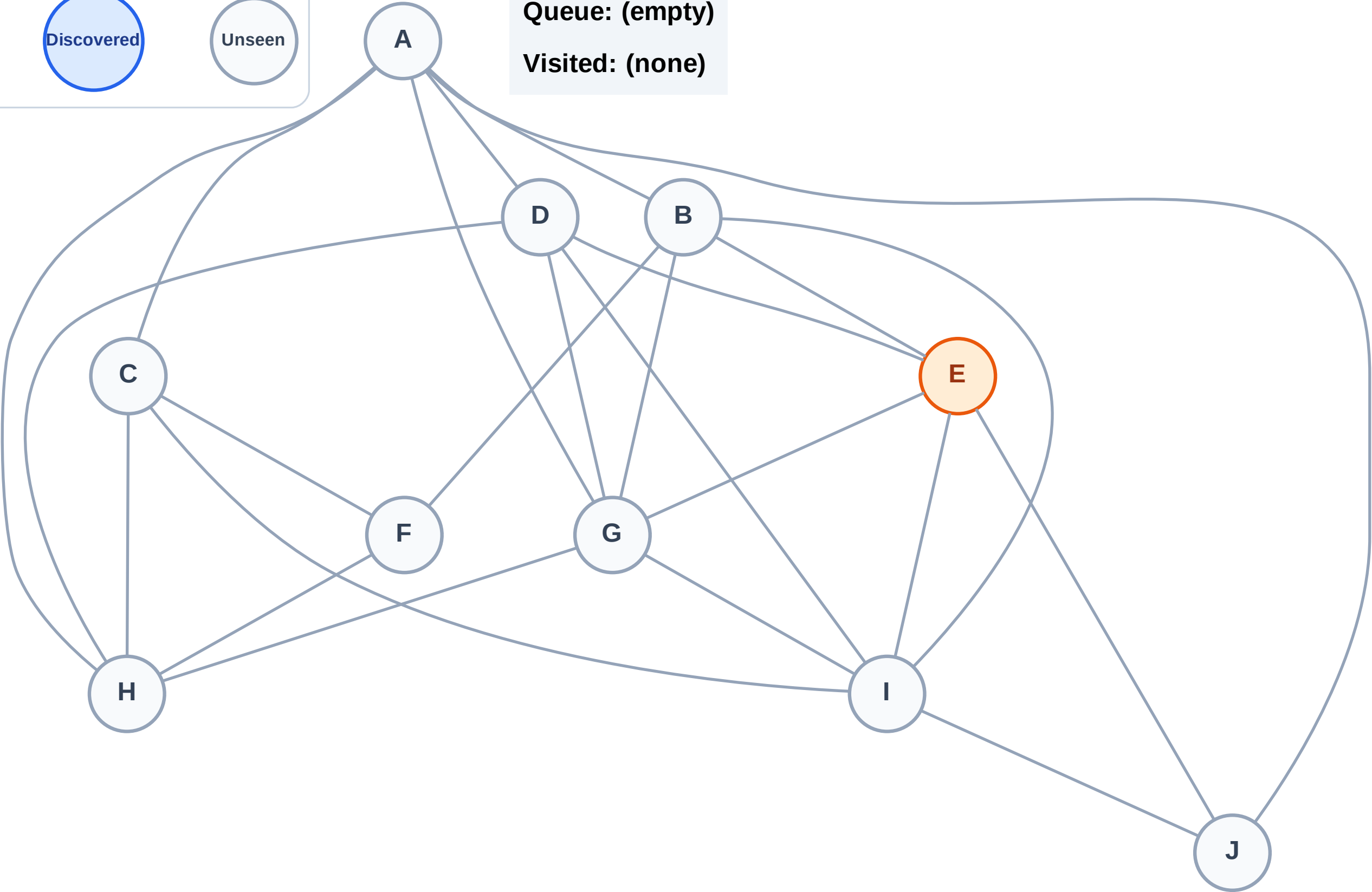
Processing

Discovered

Unseen

Queue: (empty)

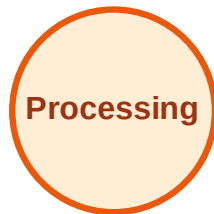
Visited: (none)



BFS Traversal

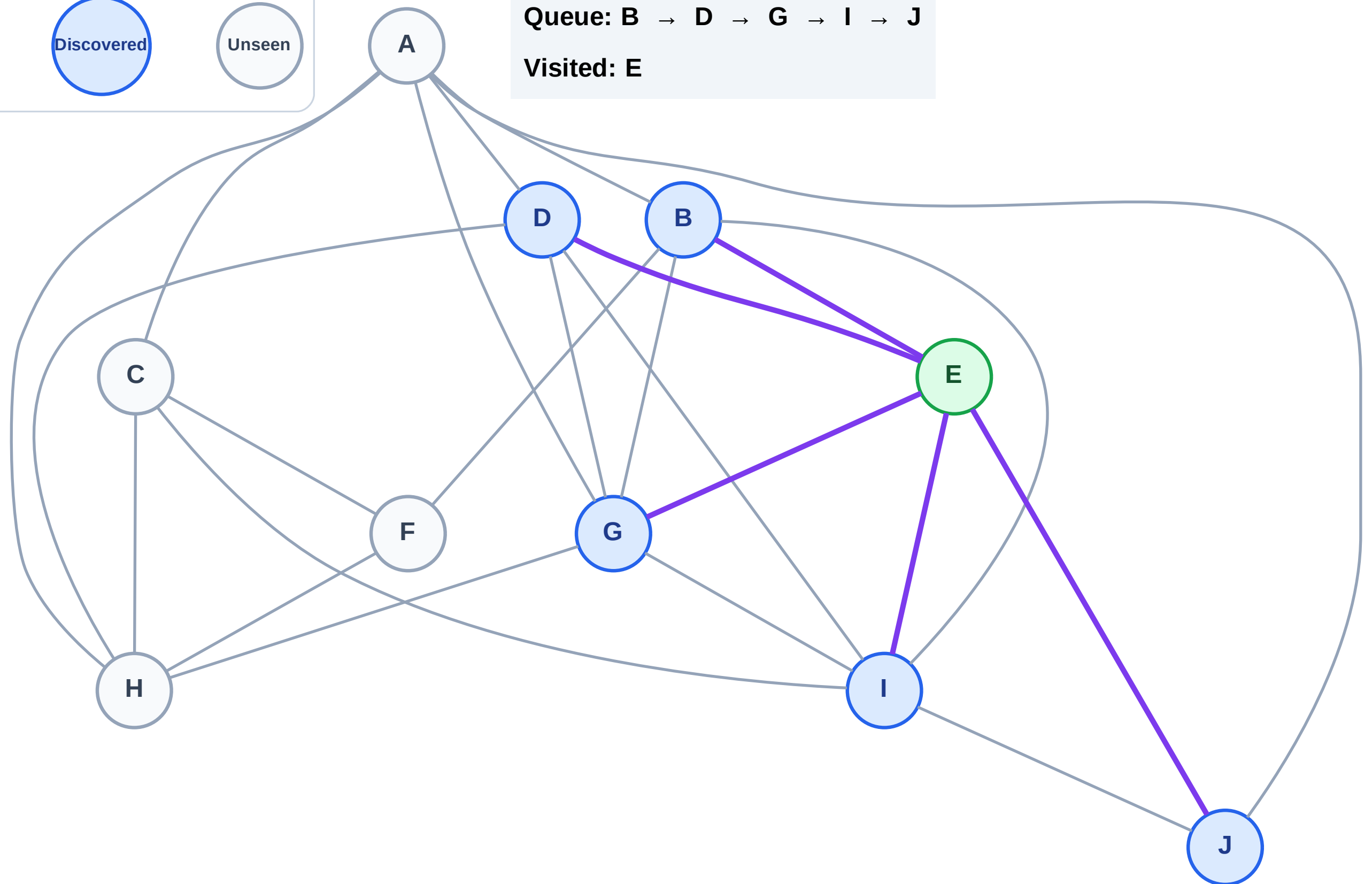
Step 3: Discover B, D, G, I, J from E

Legend



Queue: B → D → G → I → J

Visited: E



BFS Traversal

Step 4: Process node B

Legend

Complete

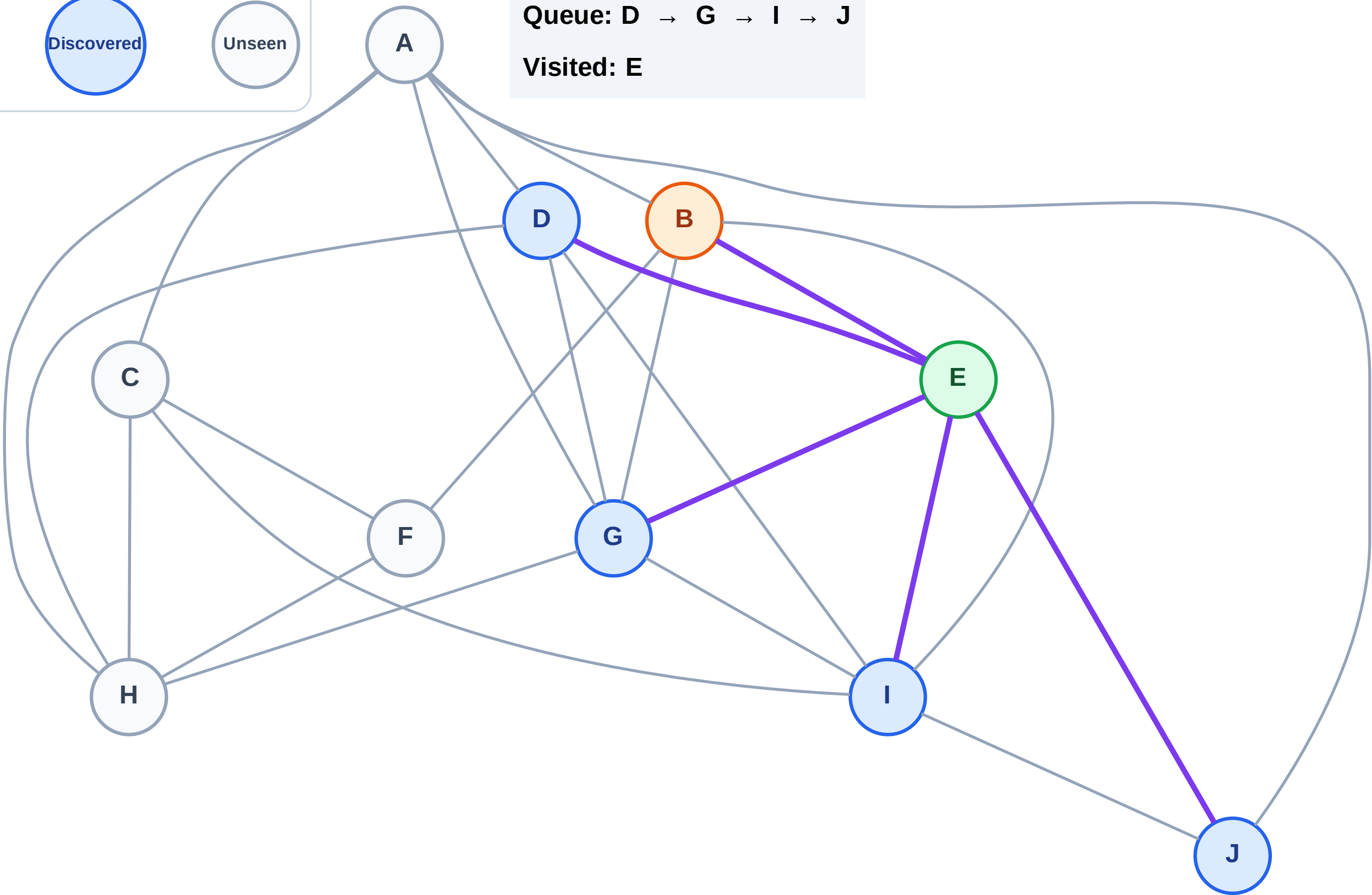
Processing

Discovered

Unseen

Queue: D → G → I → J

Visited: E



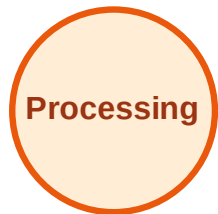
BFS Traversal

Step 5: Discover A, F from B

Legend



Complete



Processing



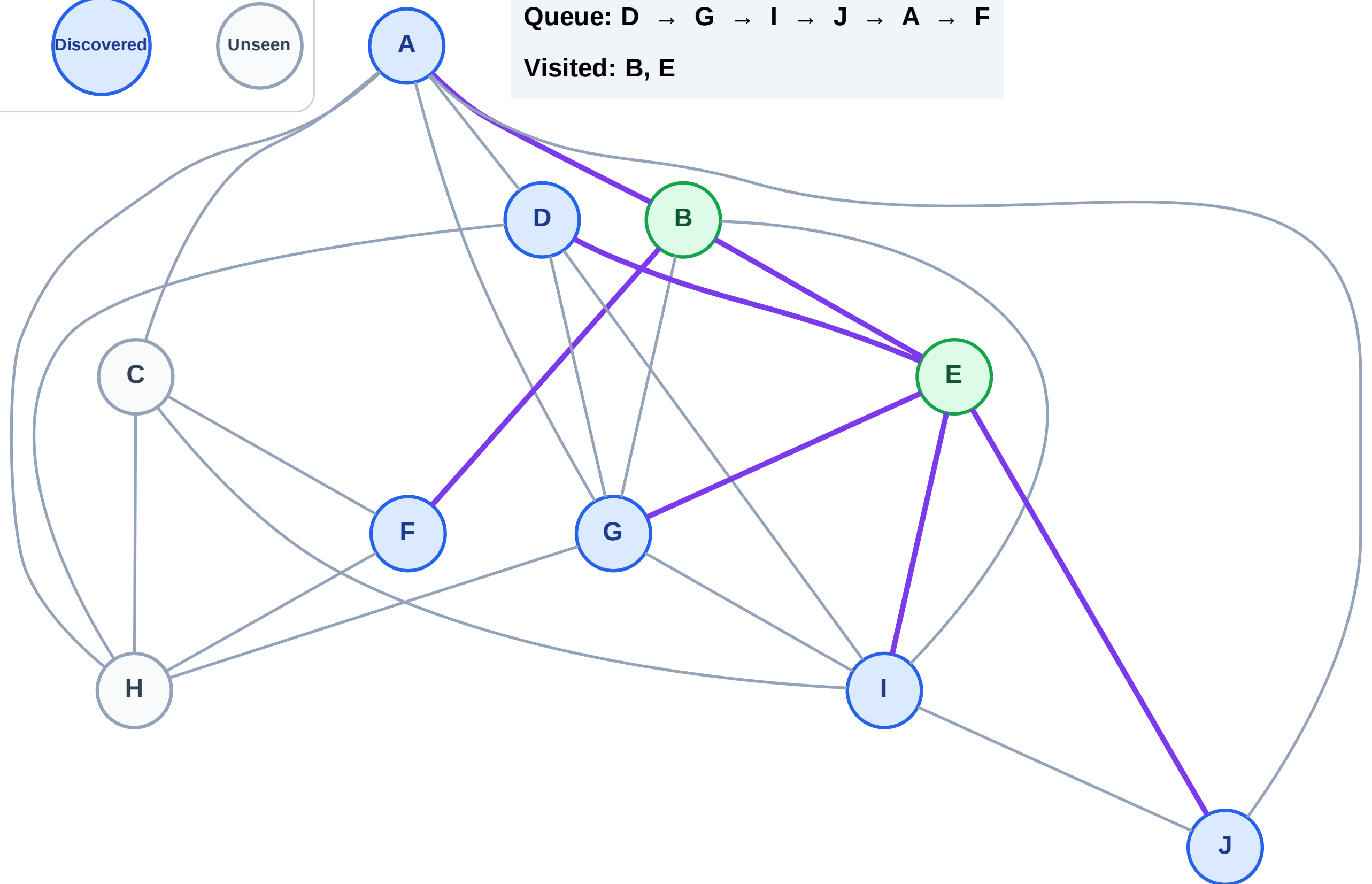
Discovered



Unseen

Queue: D → G → I → J → A → F

Visited: B, E



BFS Traversal

Step 6: Process node D

Legend

Complete

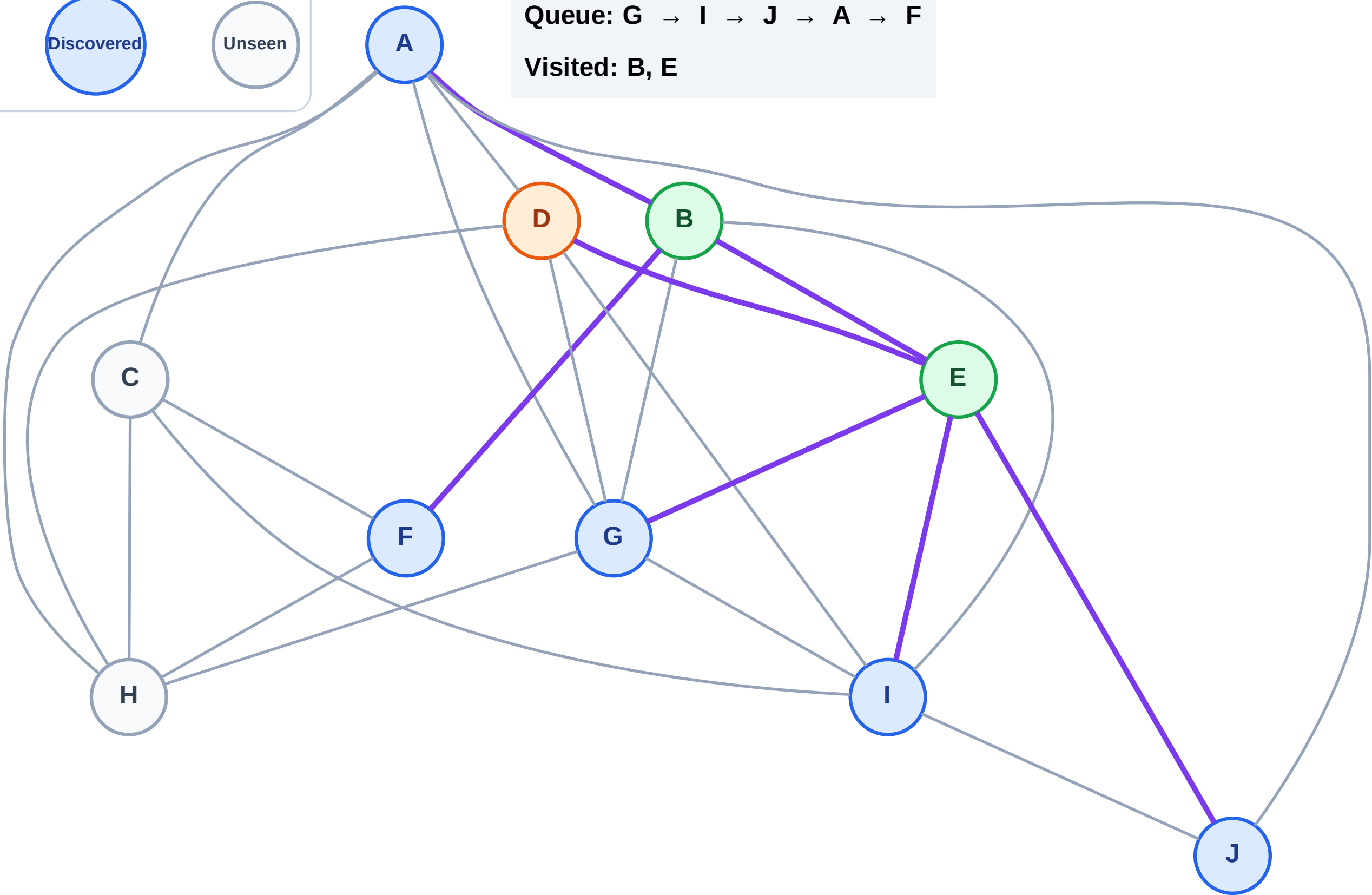
Processing

Discovered

Unseen

Queue: G → I → J → A → F

Visited: B, E



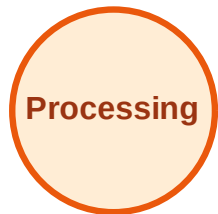
BFS Traversal

Step 7: Discover H from D

Legend



Complete



Processing



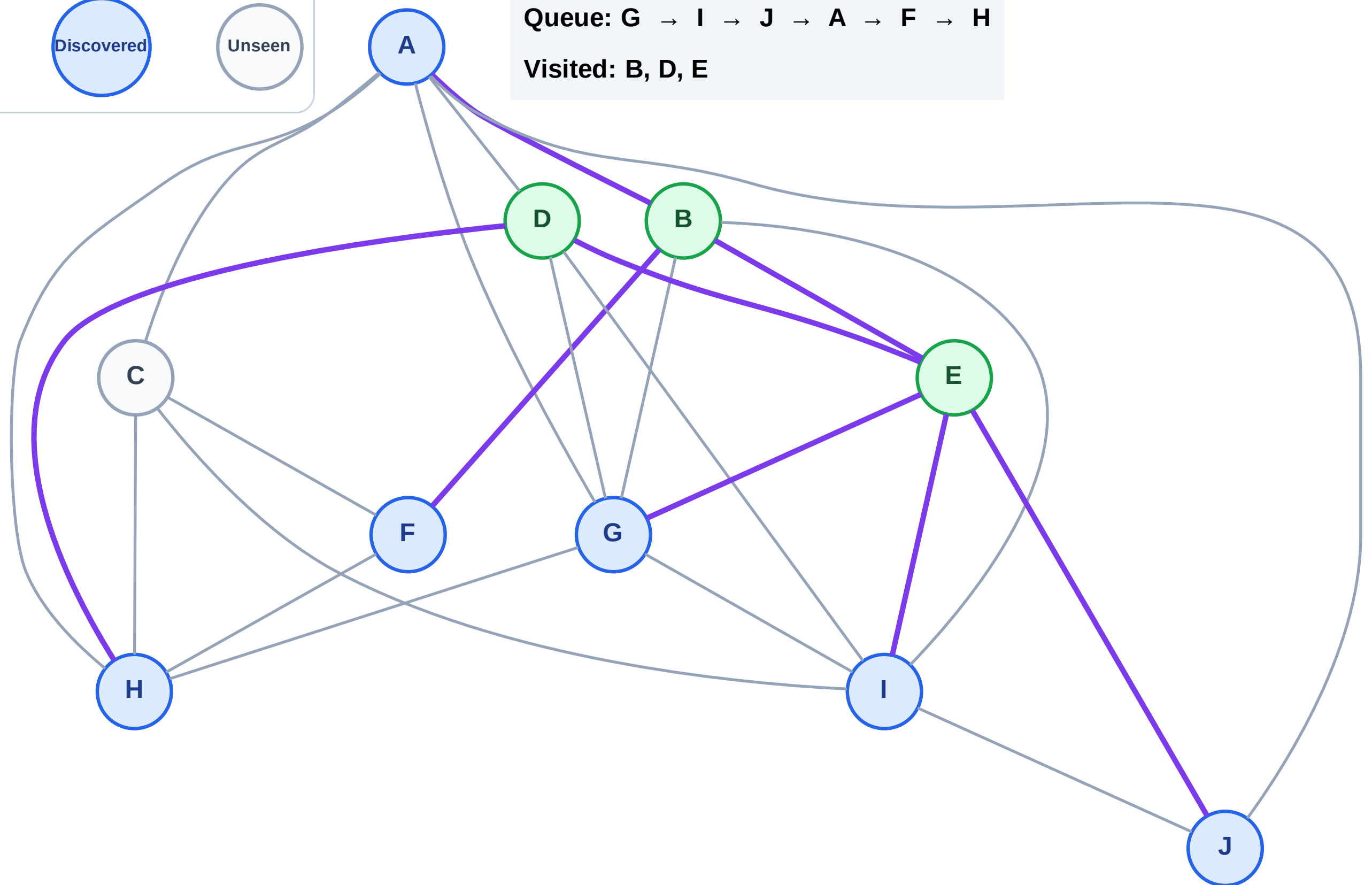
Discovered



Unseen

Queue: G → I → J → A → F → H

Visited: B, D, E



BFS Traversal

Step 8: Process node G

Legend

Complete

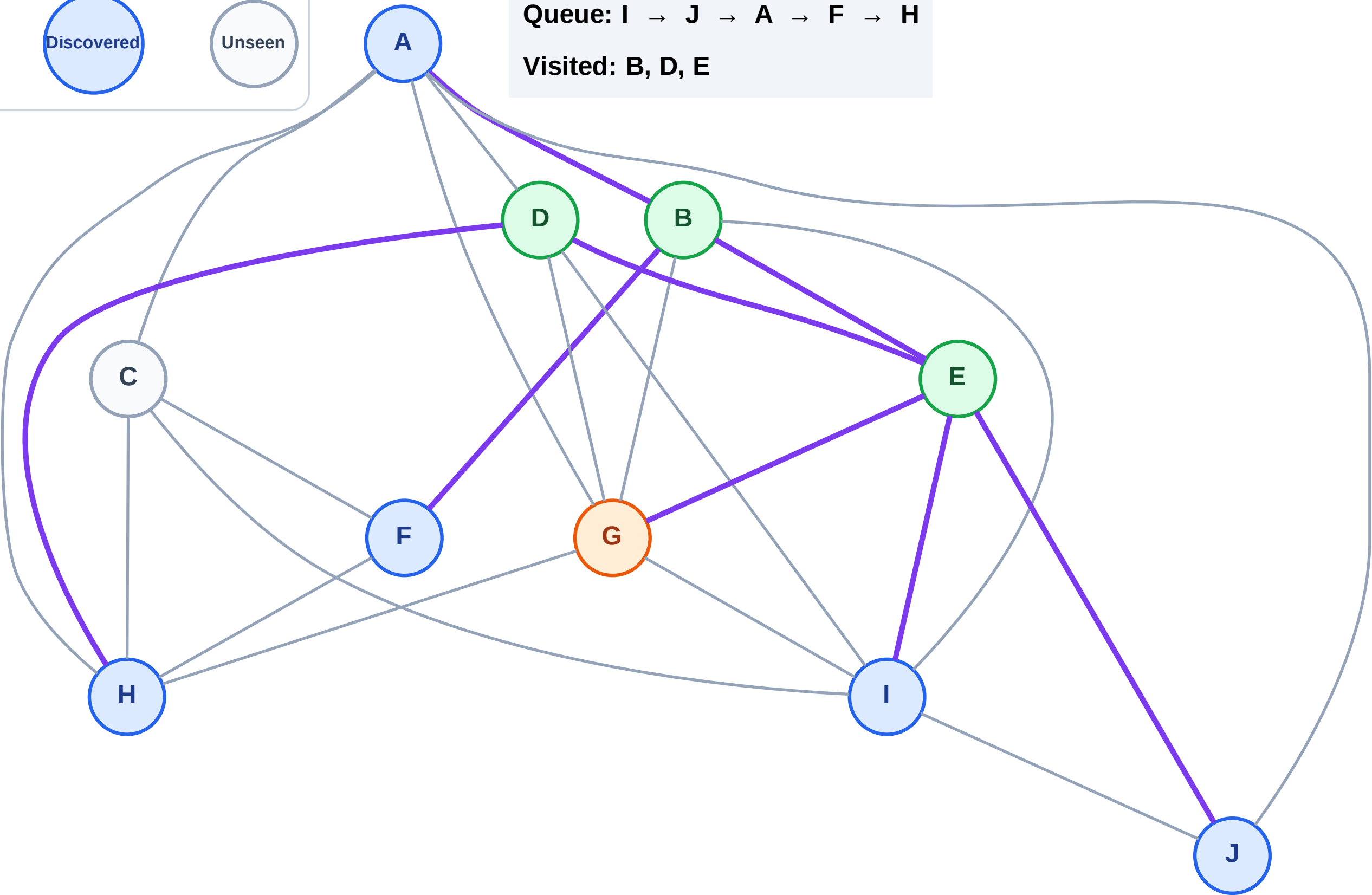
Processing

Discovered

Unseen

Queue: I → J → A → F → H

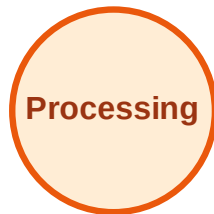
Visited: B, D, E



BFS Traversal

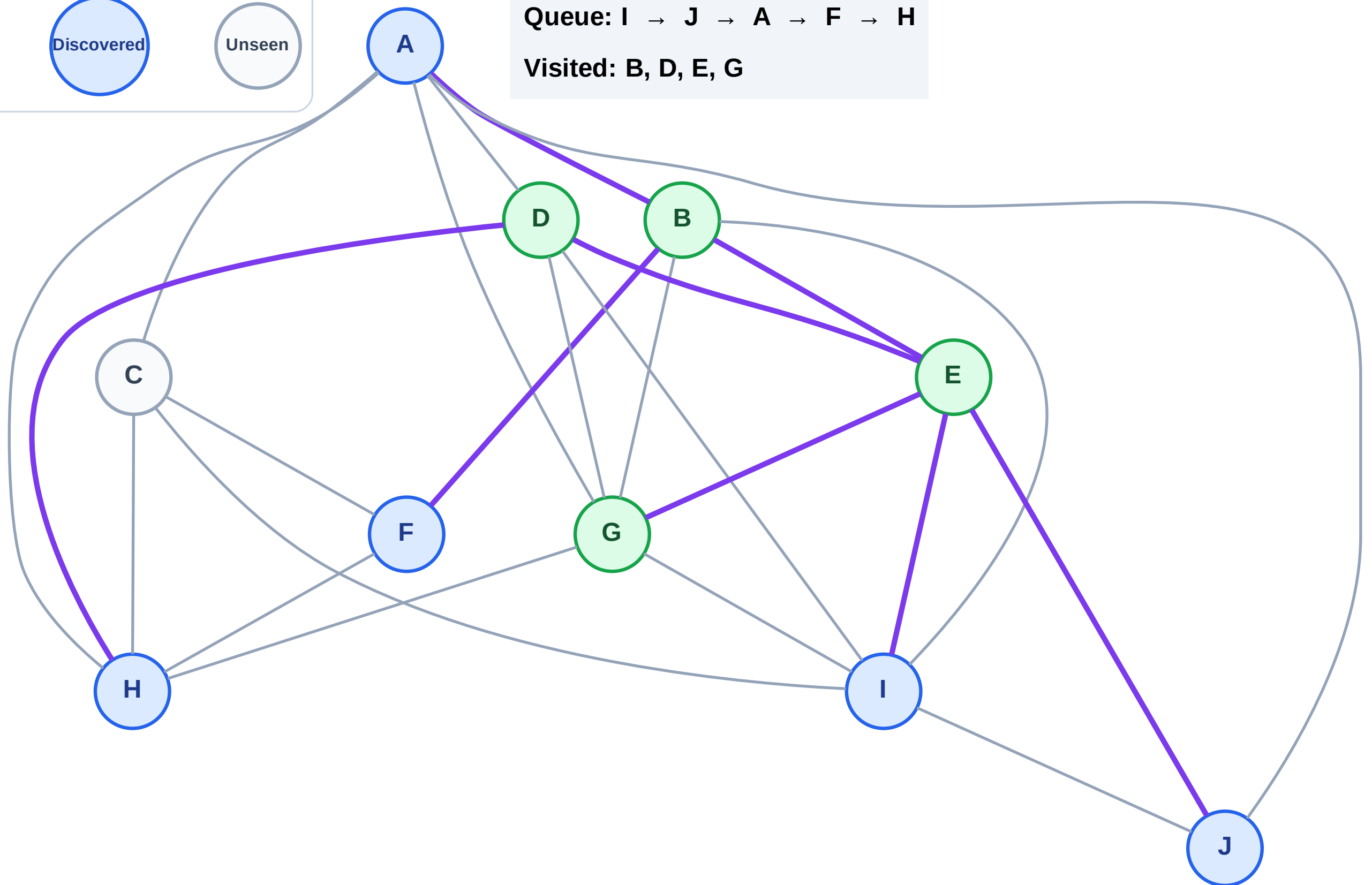
Step 9: No new neighbors from G

Legend



Queue: I → J → A → F → H

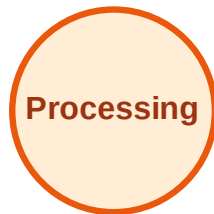
Visited: B, D, E, G



BFS Traversal

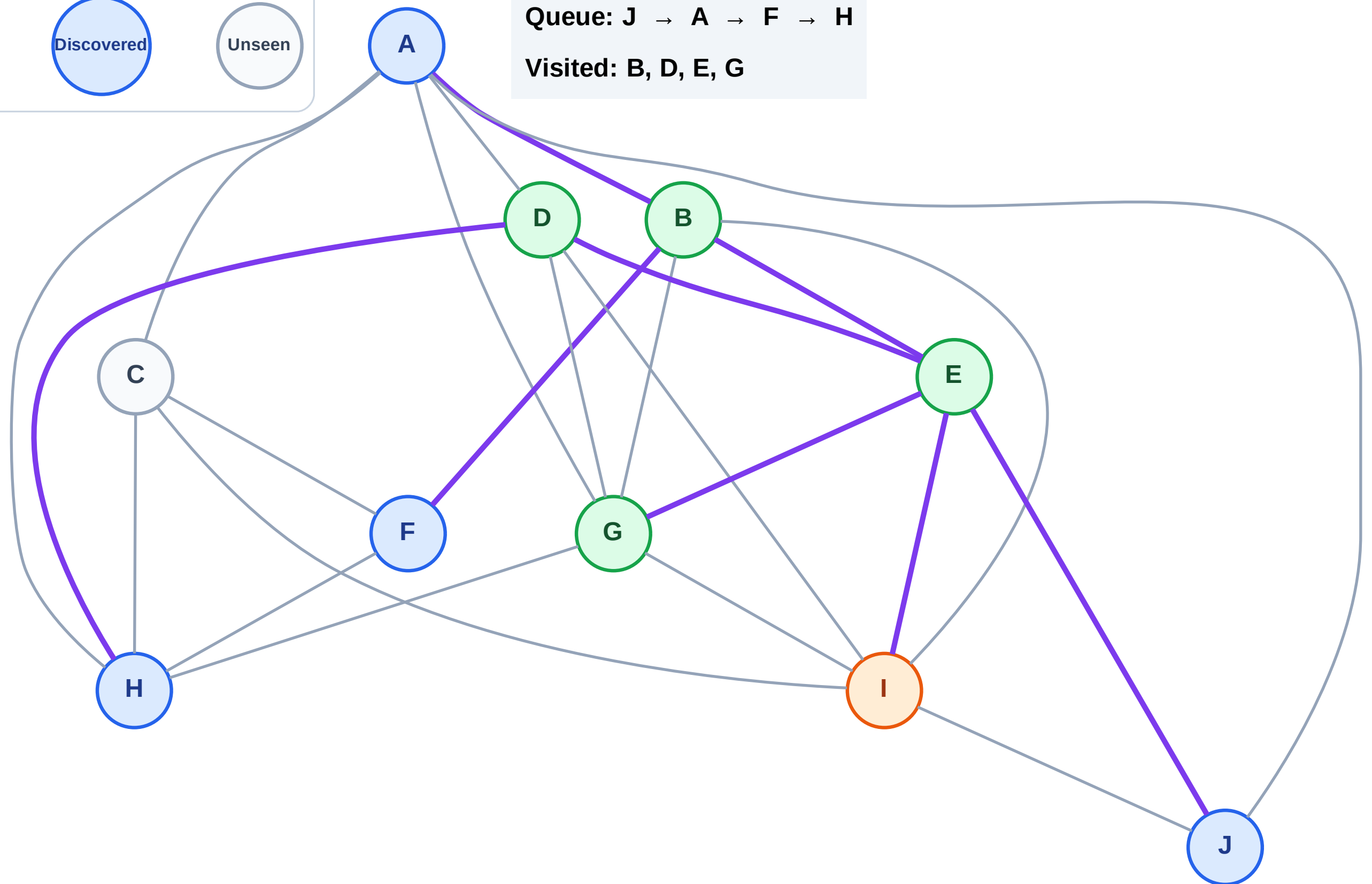
Step 10: Process node I

Legend



Queue: J → A → F → H

Visited: B, D, E, G



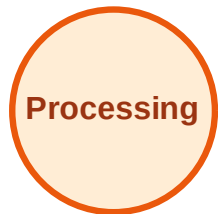
BFS Traversal

Step 11: Discover C from I

Legend



Complete



Processing



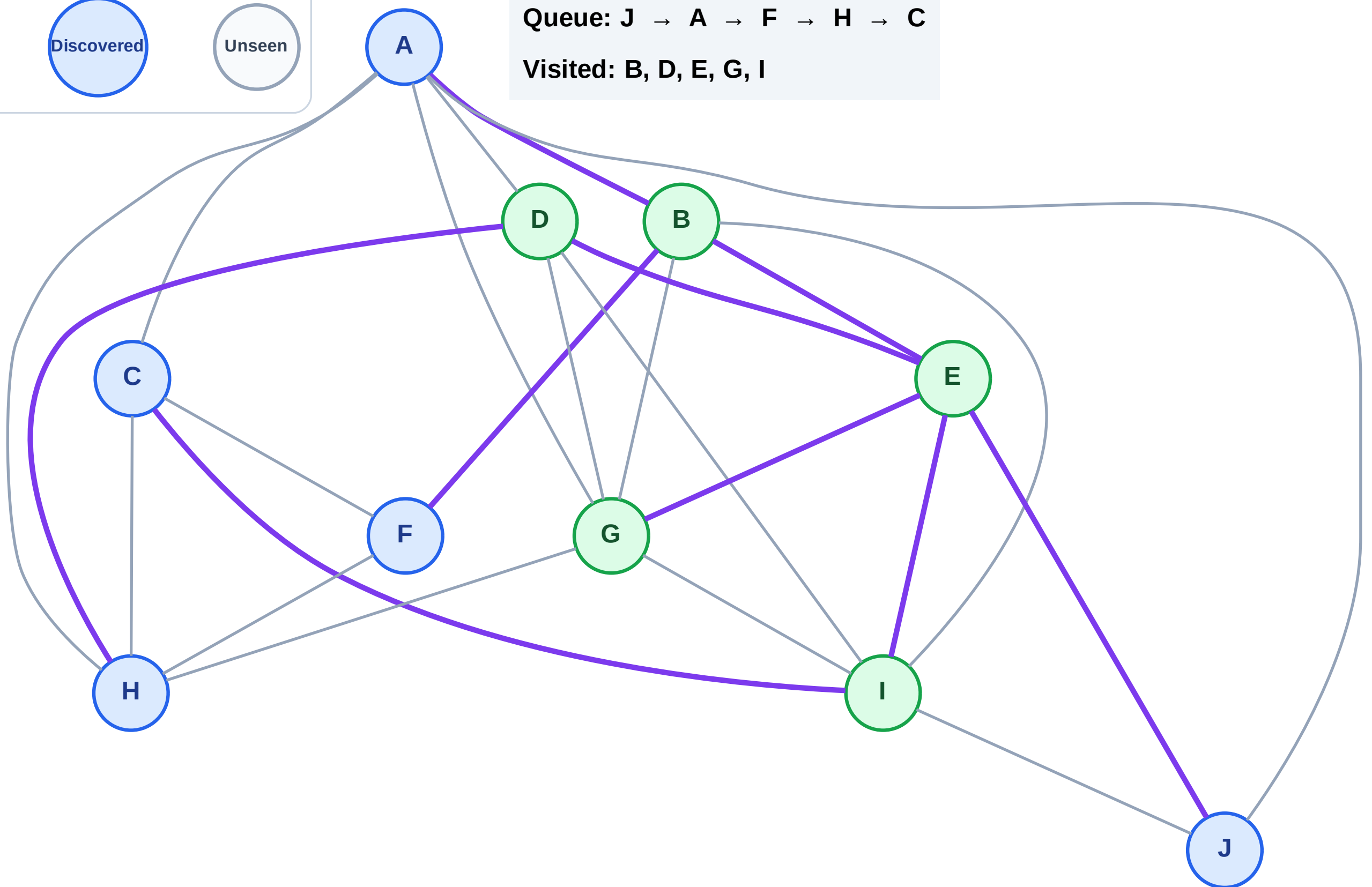
Discovered



Unseen

Queue: J → A → F → H → C

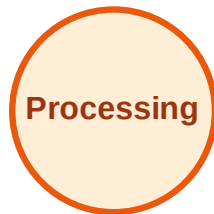
Visited: B, D, E, G, I



BFS Traversal

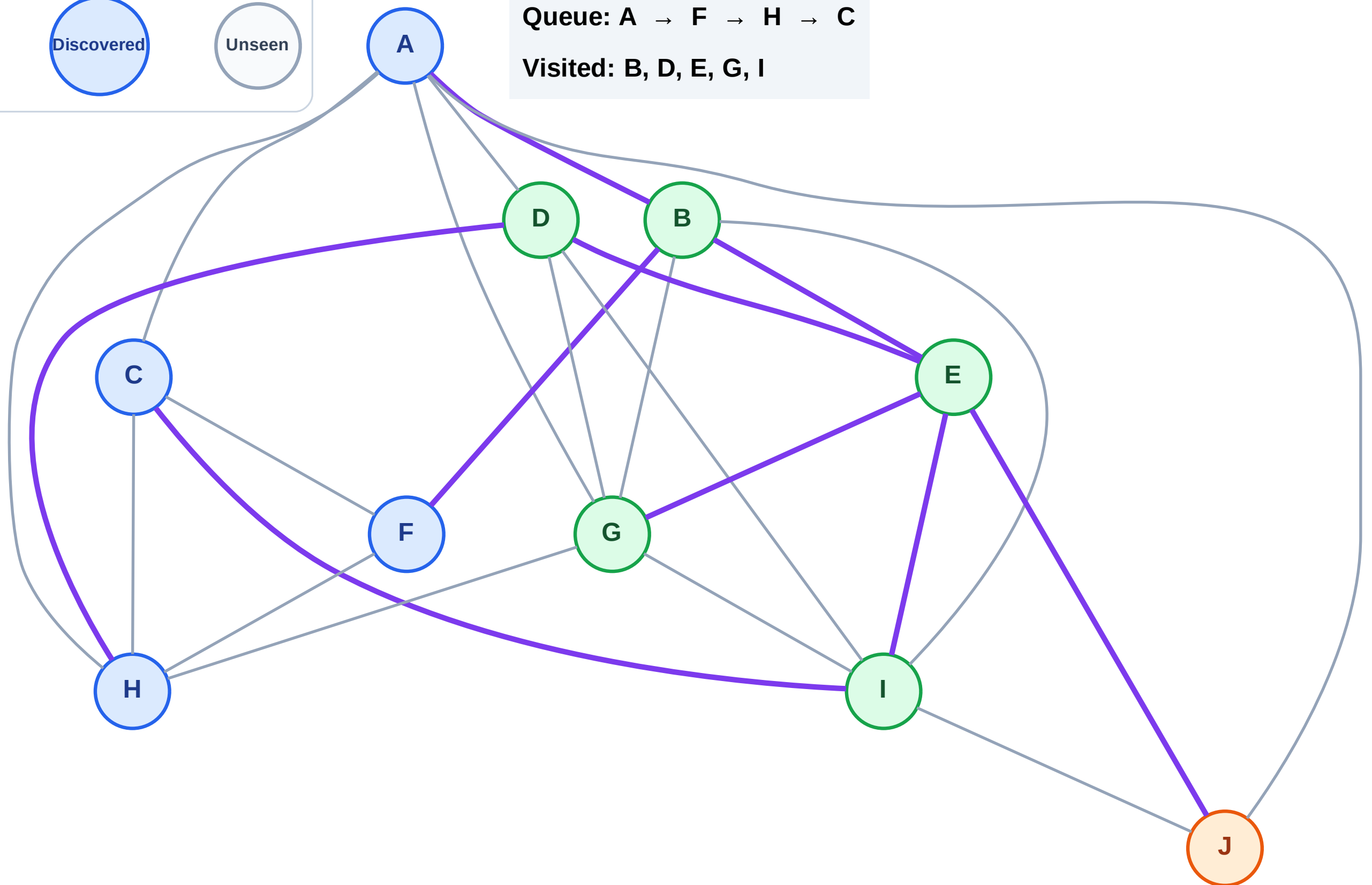
Step 12: Process node J

Legend



Queue: A → F → H → C

Visited: B, D, E, G, I



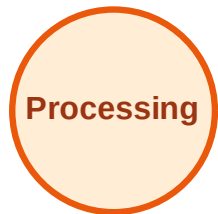
BFS Traversal

Step 13: No new neighbors from J

Legend



Complete



Processing



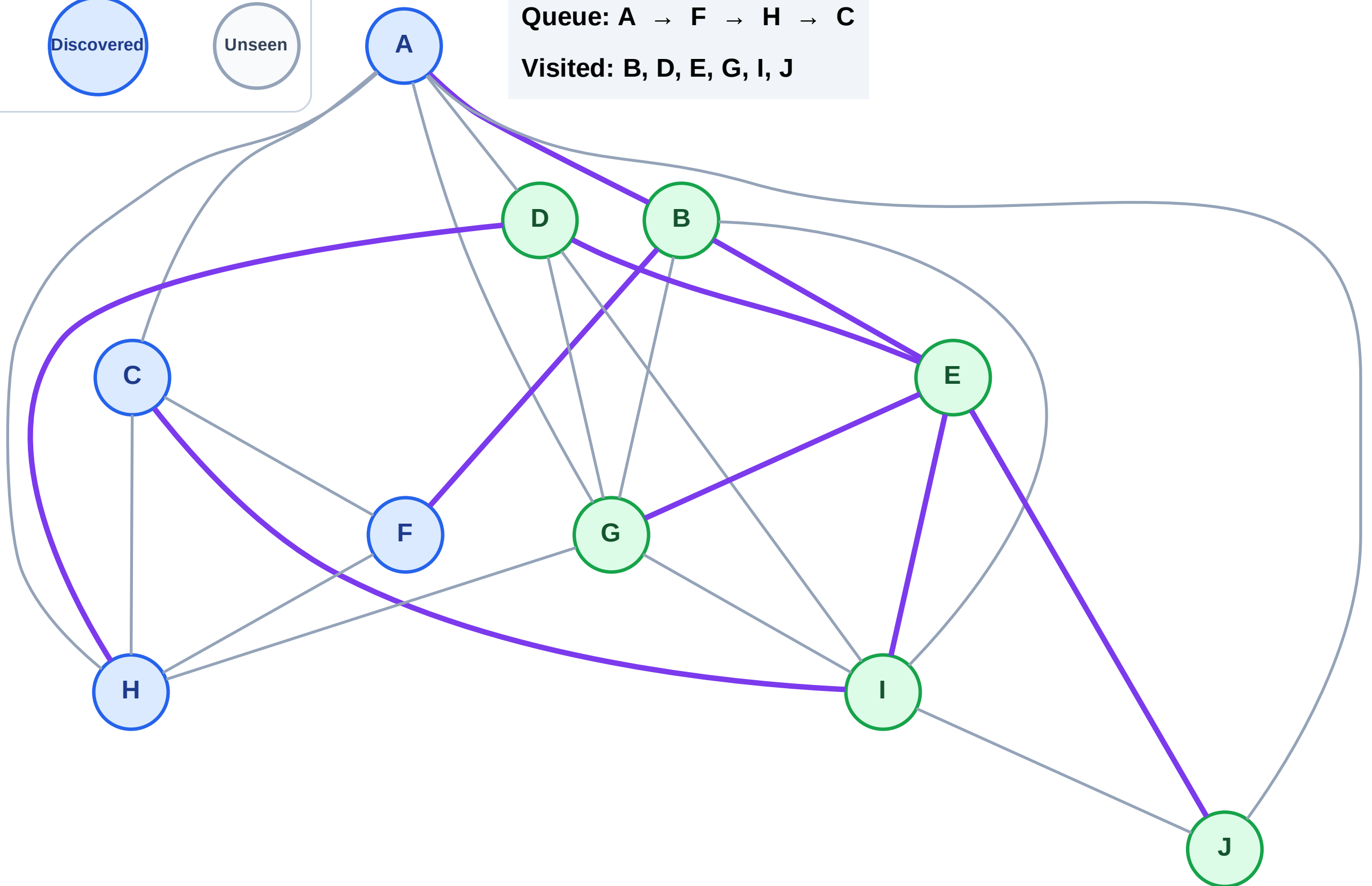
Discovered



Unseen

Queue: A → F → H → C

Visited: B, D, E, G, I, J



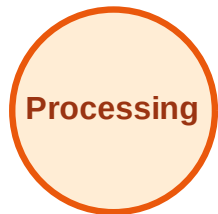
BFS Traversal

Step 14: Process node A

Legend



Complete



Processing



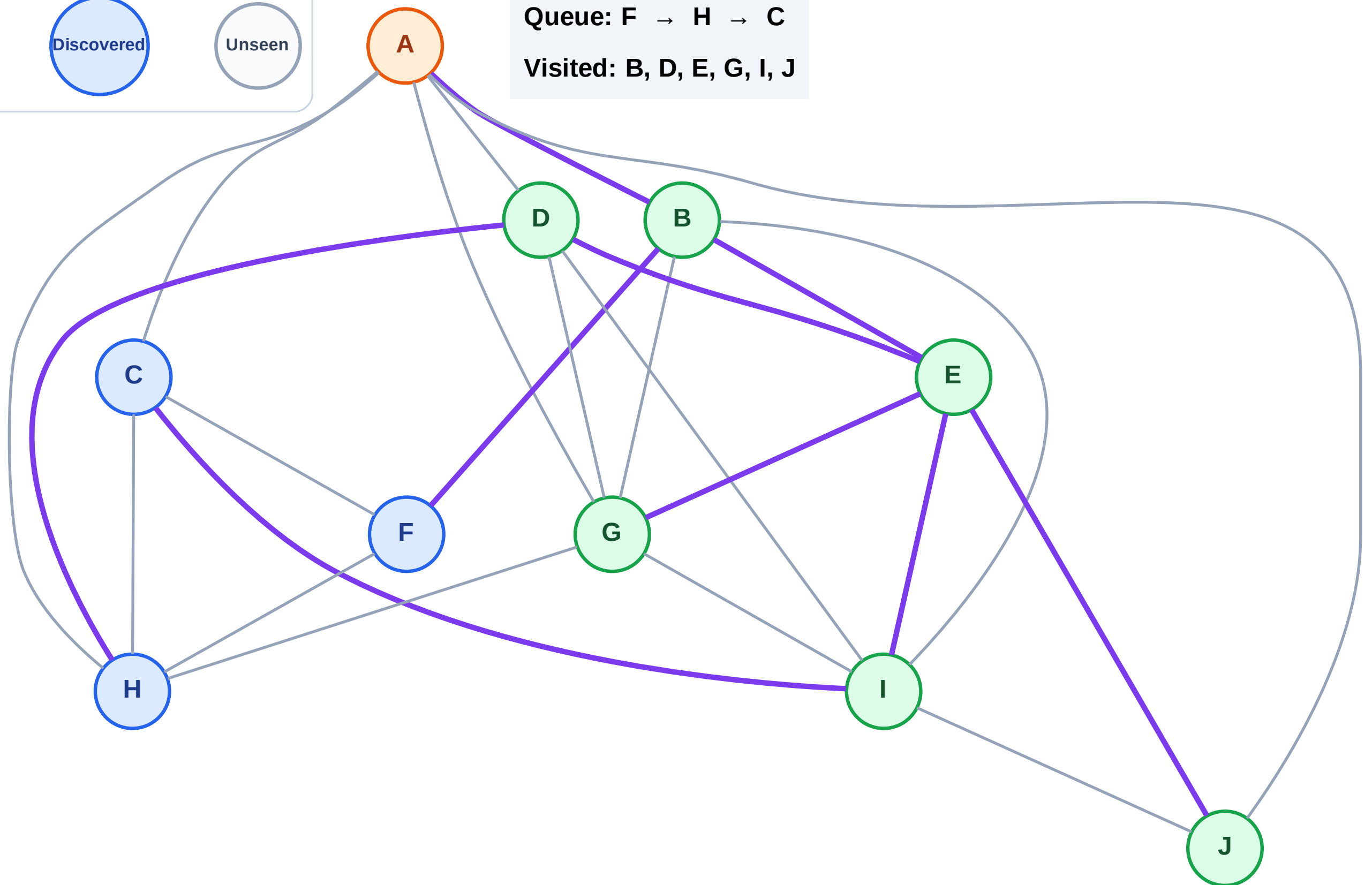
Discovered



Unseen

Queue: F → H → C

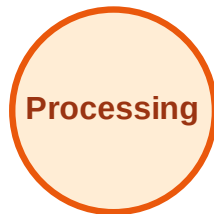
Visited: B, D, E, G, I, J



BFS Traversal

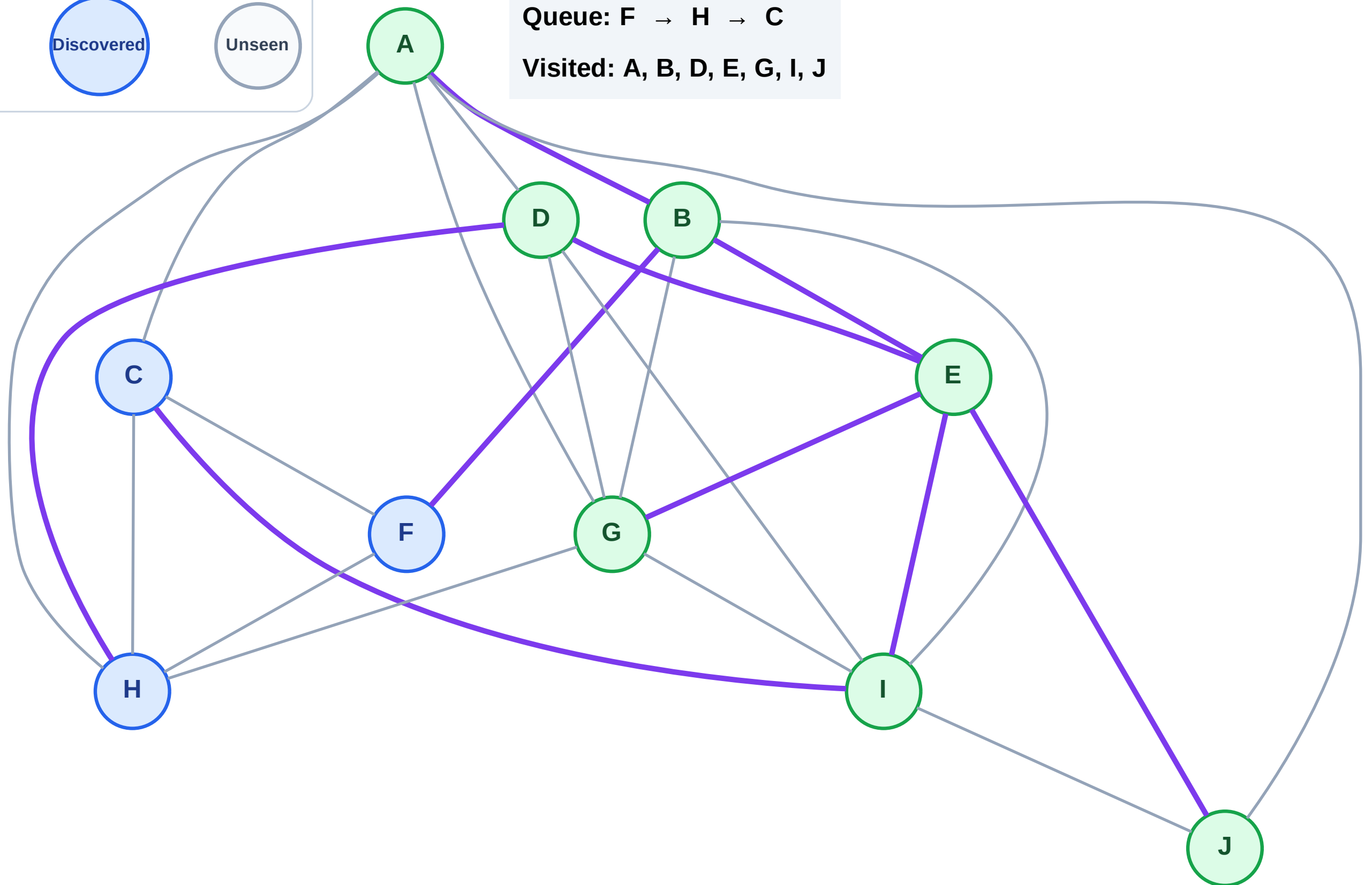
Step 15: No new neighbors from A

Legend



Queue: F → H → C

Visited: A, B, D, E, G, I, J



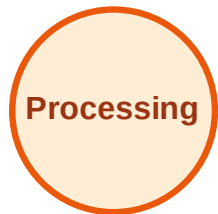
BFS Traversal

Step 16: Process node F

Legend



Complete



Processing



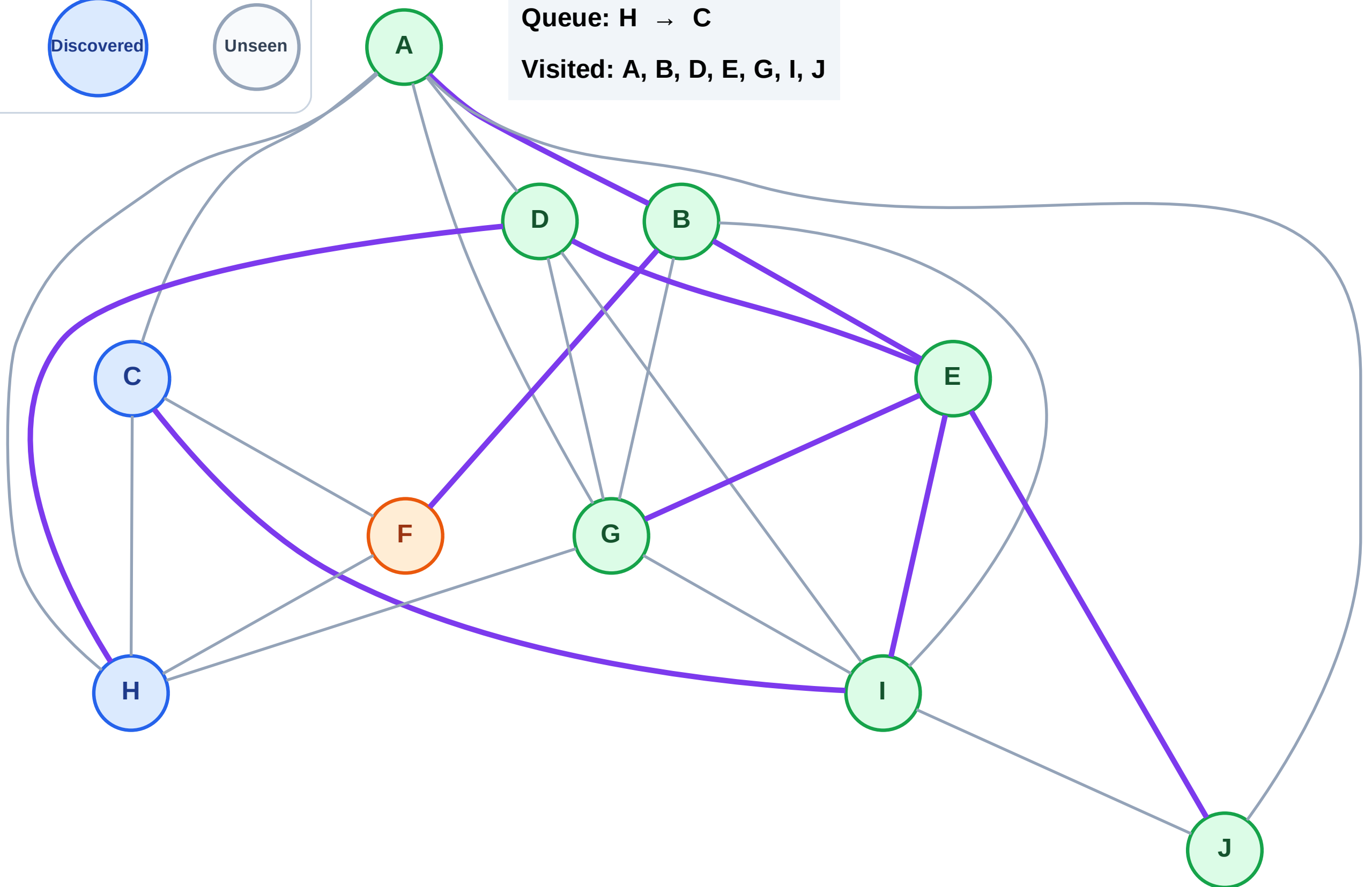
Discovered



Unseen

Queue: H → C

Visited: A, B, D, E, G, I, J



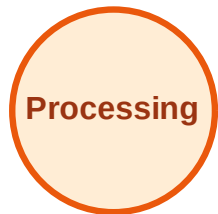
BFS Traversal

Step 17: No new neighbors from F

Legend



Complete



Processing



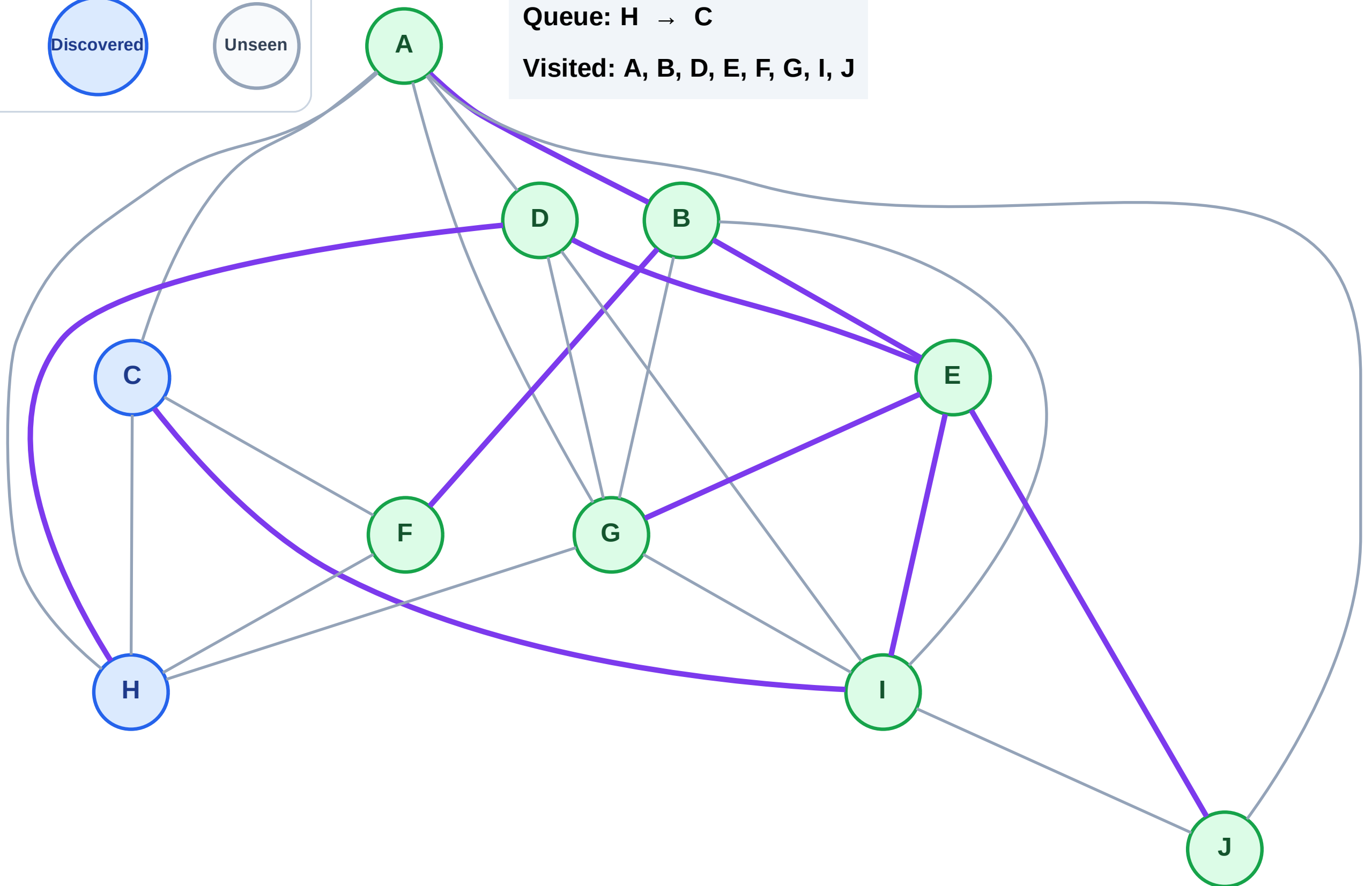
Discovered



Unseen

Queue: H → C

Visited: A, B, D, E, F, G, I, J



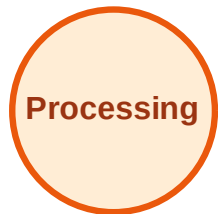
BFS Traversal

Step 18: Process node H

Legend



Complete



Processing



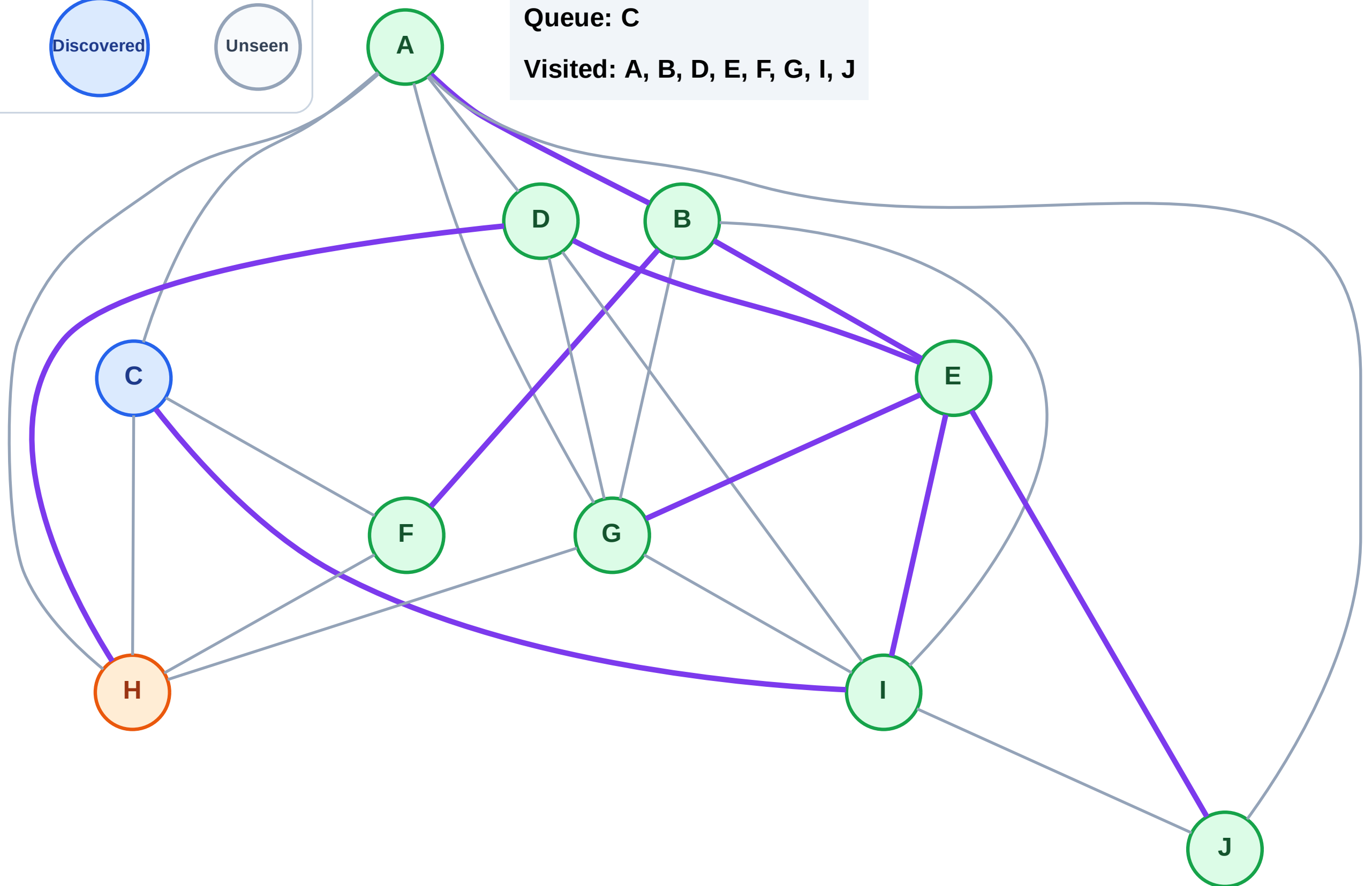
Discovered



Unseen

Queue: C

Visited: A, B, D, E, F, G, I, J



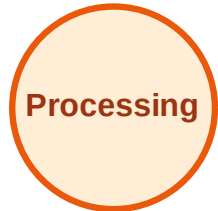
BFS Traversal

Step 19: No new neighbors from H

Legend



Complete



Processing



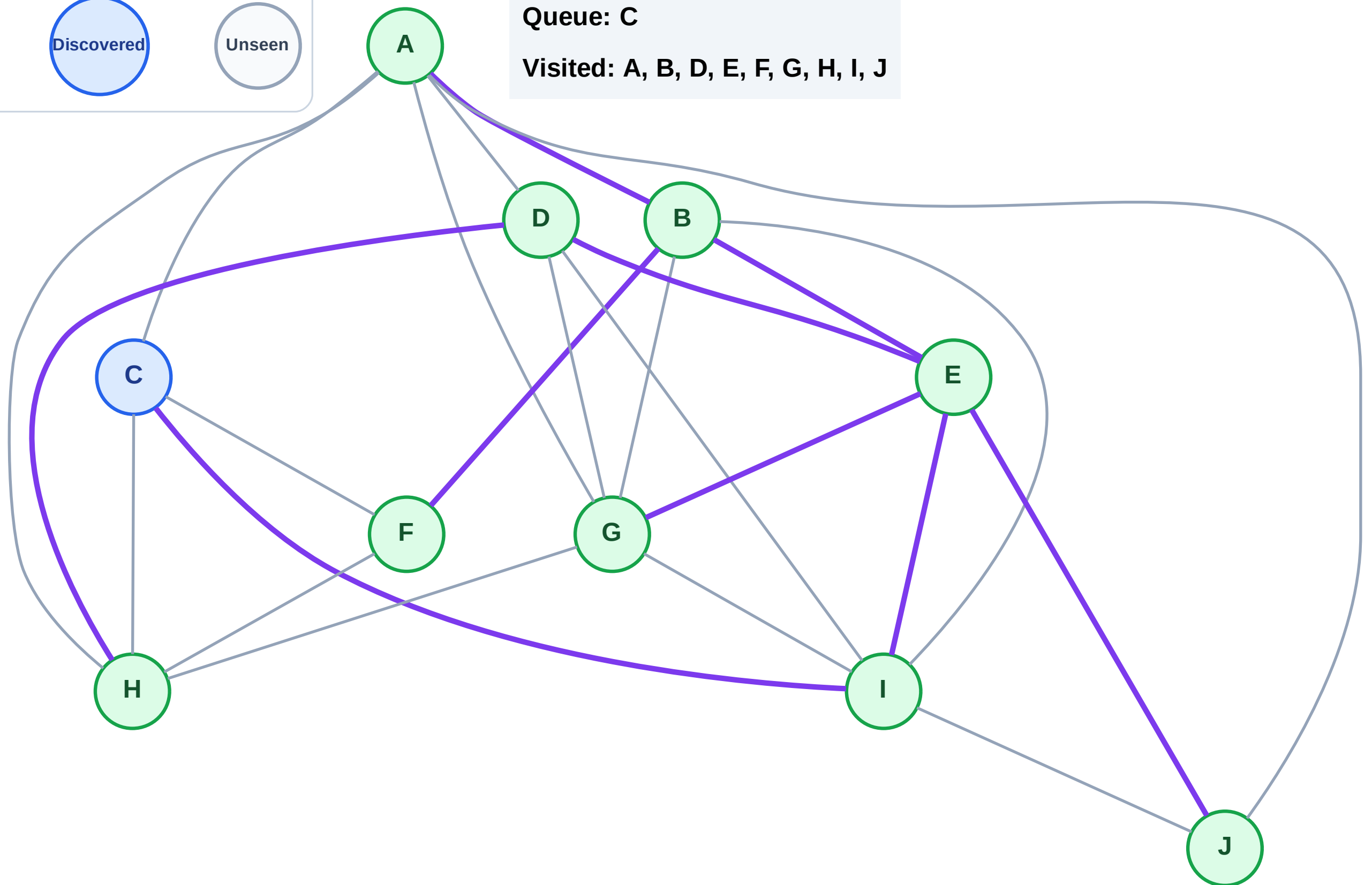
Discovered



Unseen

Queue: C

Visited: A, B, D, E, F, G, H, I, J



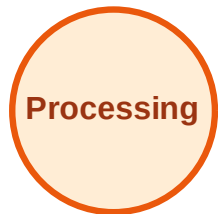
BFS Traversal

Step 20: Process node C

Legend



Complete



Processing



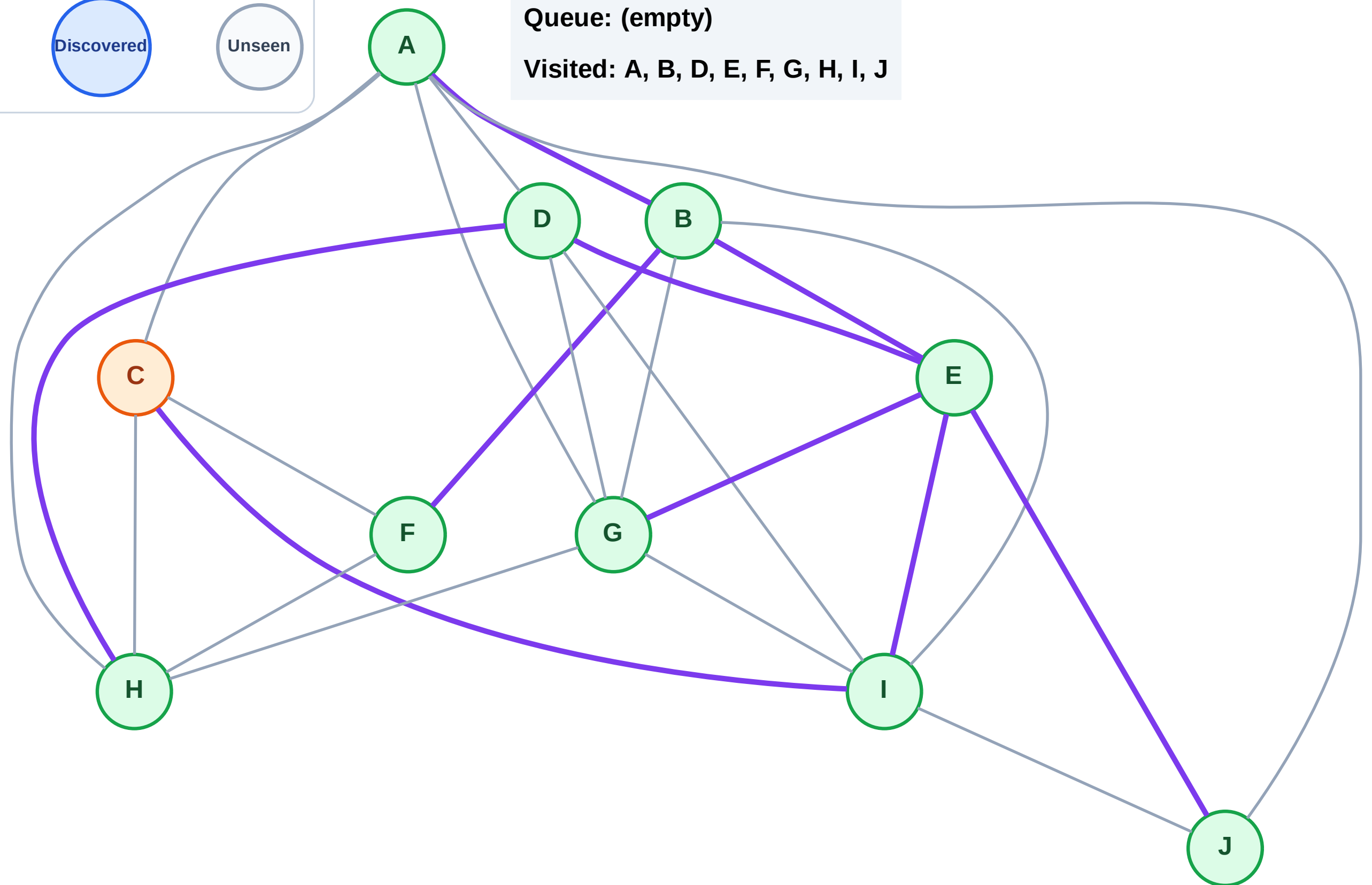
Discovered



Unseen

Queue: (empty)

Visited: A, B, D, E, F, G, H, I, J



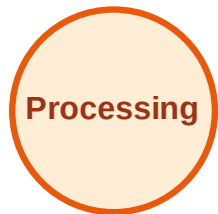
BFS Traversal

Step 21: No new neighbors from C

Legend



Complete



Processing



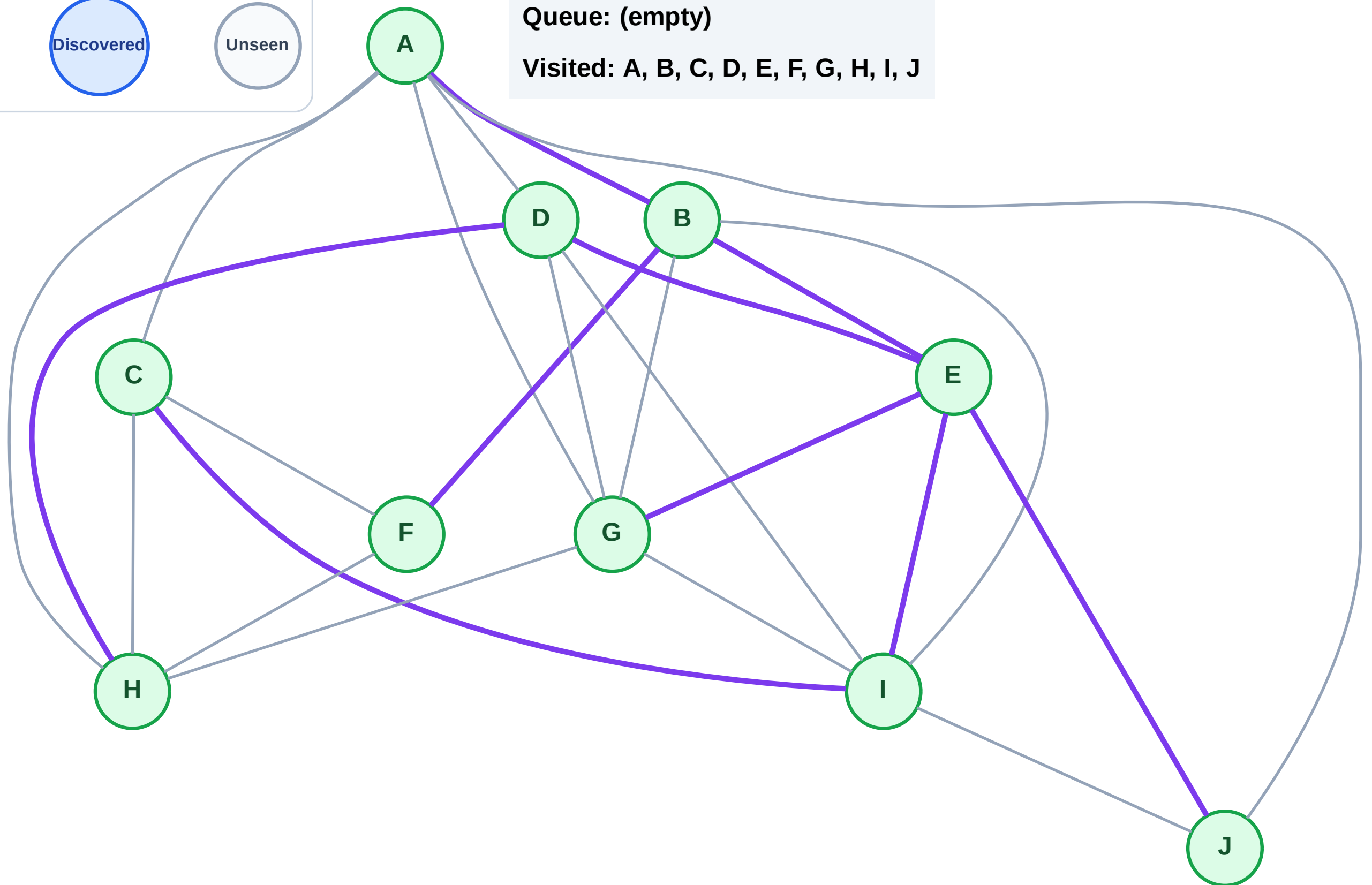
Discovered



Unseen

Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

Step 1: Start at E

Legend

Complete

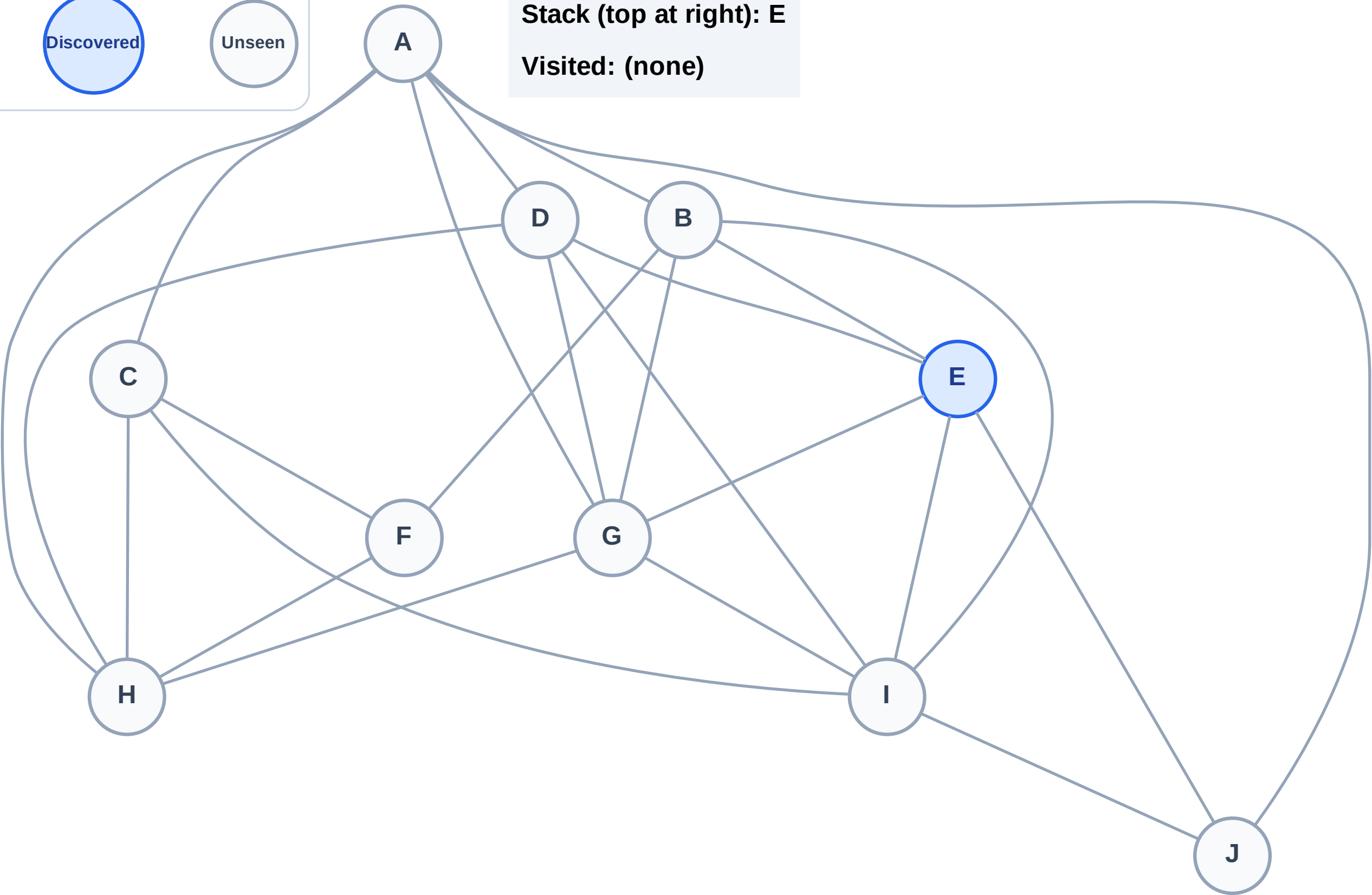
Processing

Discovered

Unseen

Stack (top at right): E

Visited: (none)



DFS Traversal

Step 2: Process node E

Legend

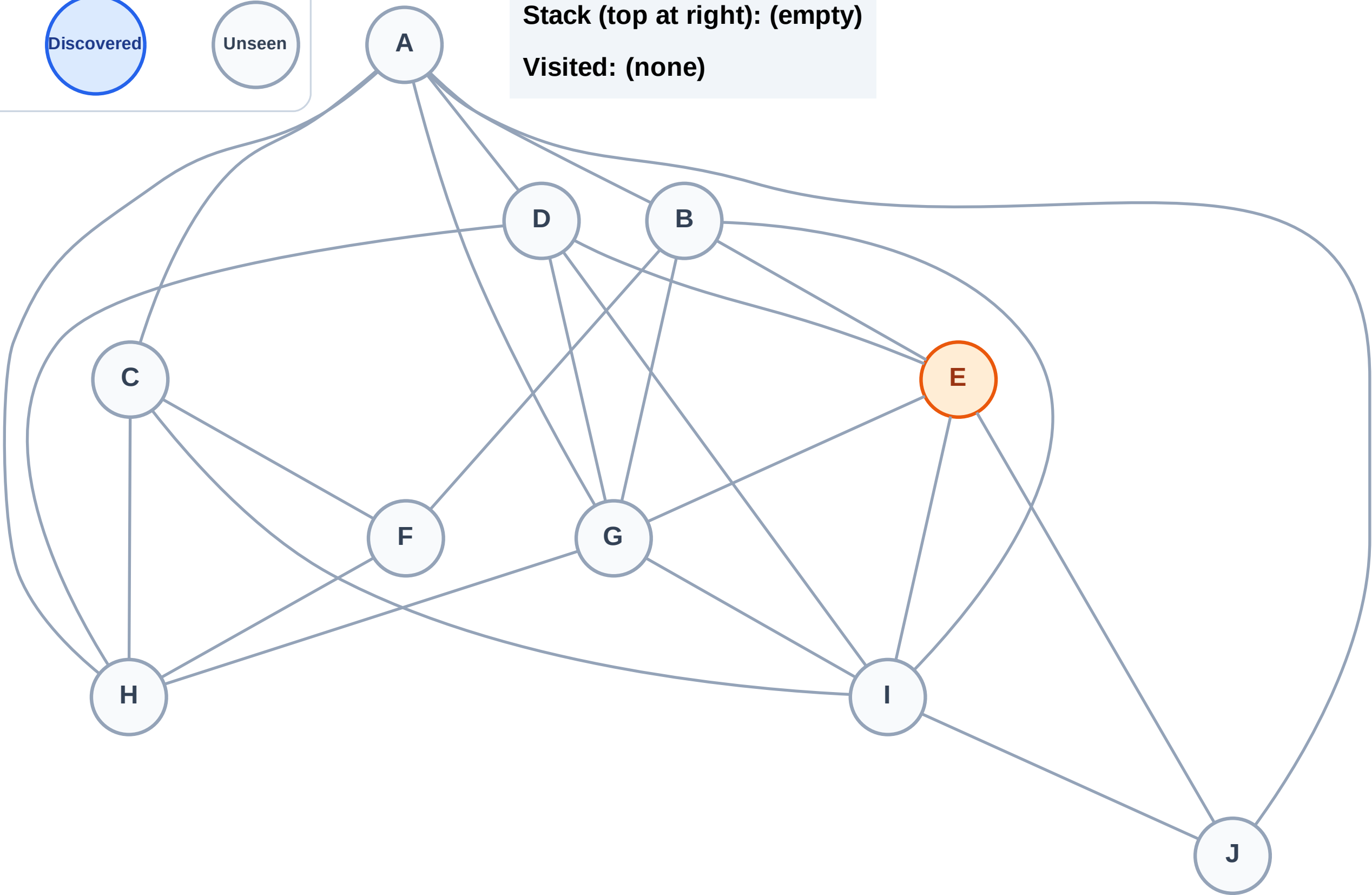
Complete

Processing

Discovered

Unseen

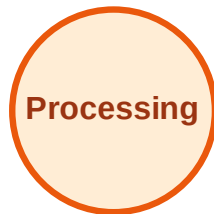
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

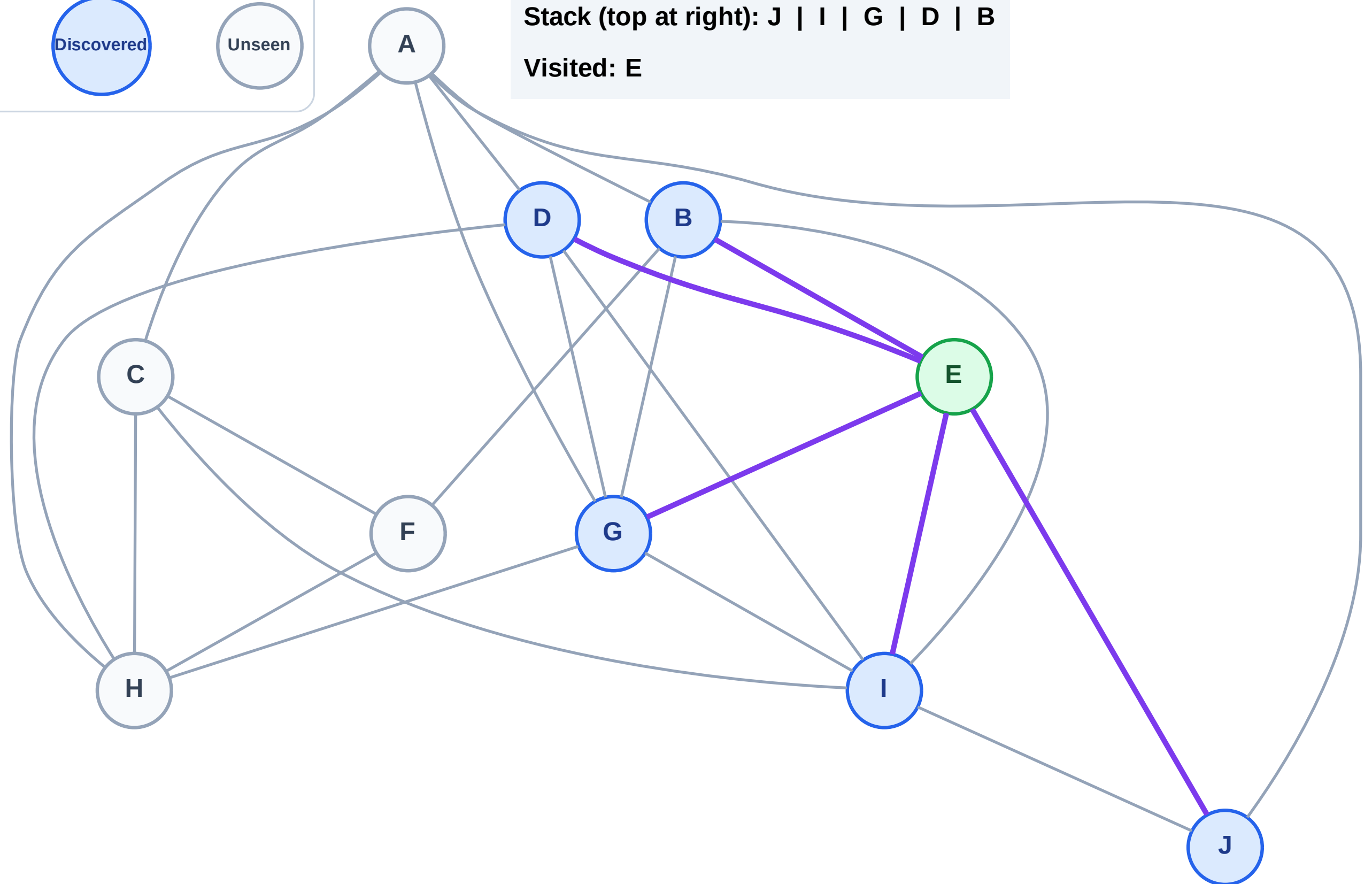
Step 3: Discover J, I, G, D, B from E

Legend



Stack (top at right): J | I | G | D | B

Visited: E



DFS Traversal

Step 4: Process node B

Legend

Complete

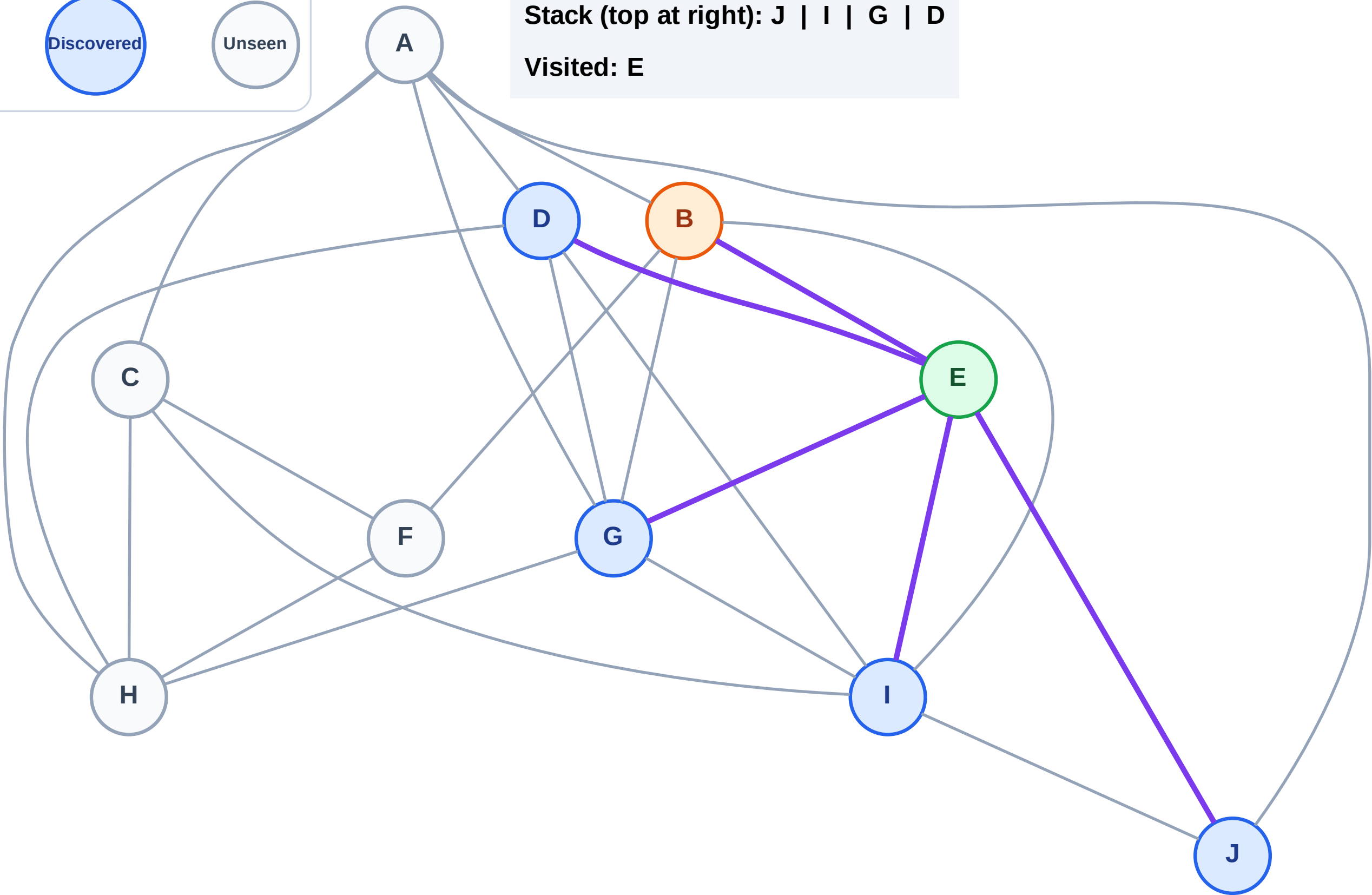
Processing

Discovered

Unseen

Stack (top at right): J | I | G | D

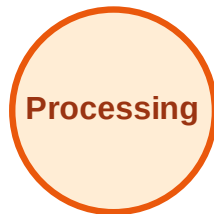
Visited: E



DFS Traversal

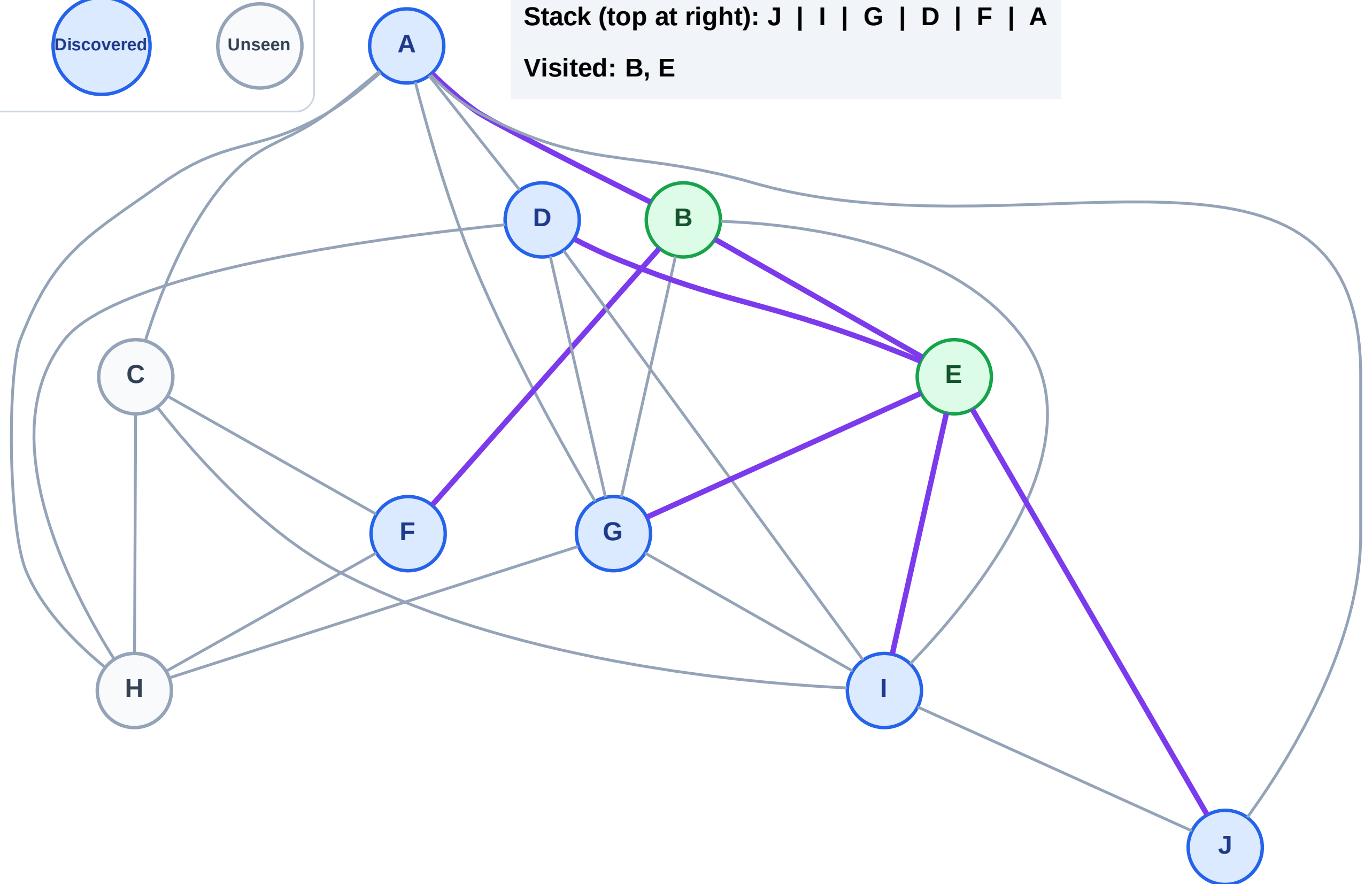
Step 5: Discover F, A from B

Legend



Stack (top at right): J | I | G | D | F | A

Visited: B, E



DFS Traversal

Step 6: Process node A

Legend

Complete

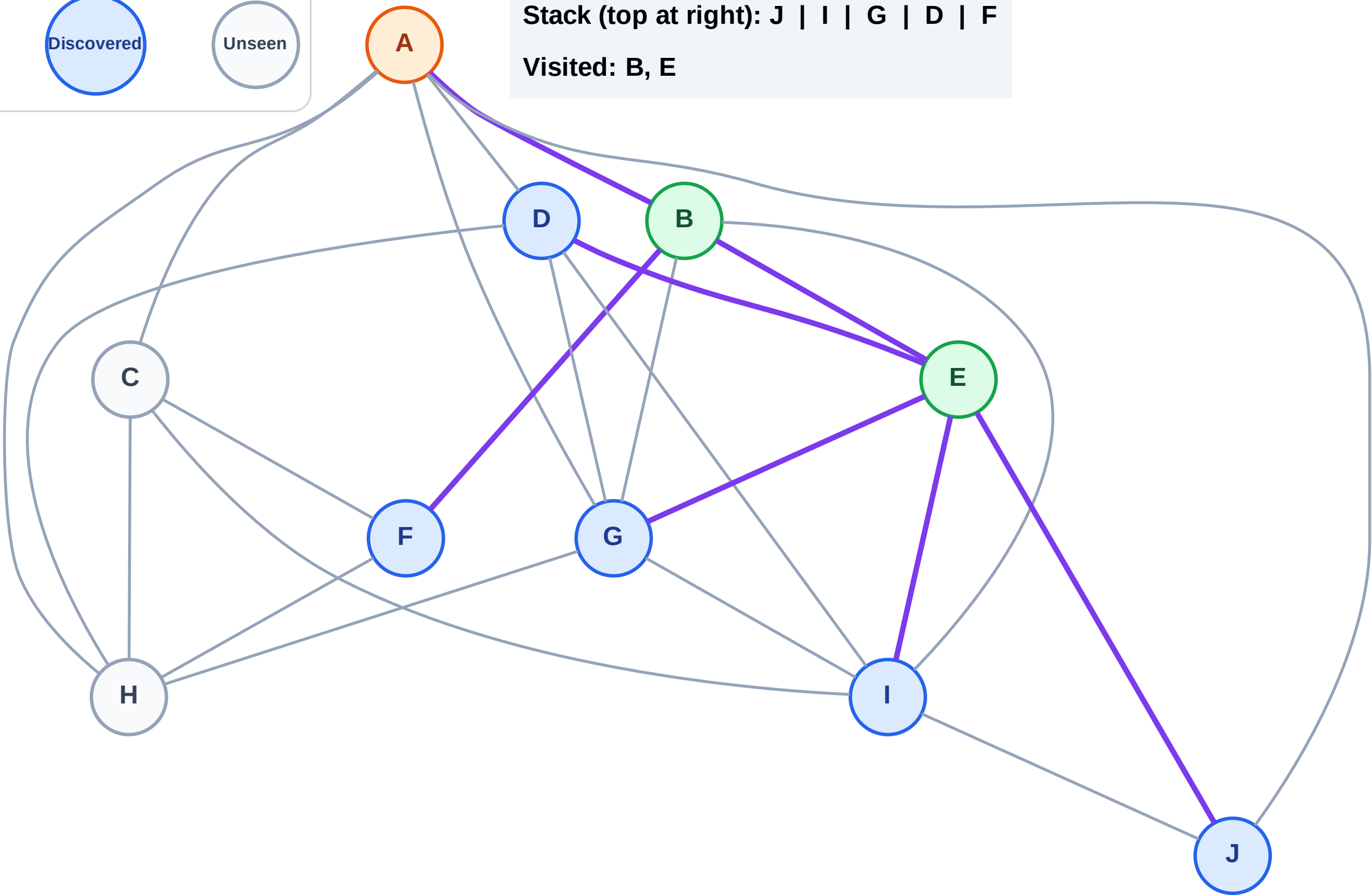
Processing

Discovered

Unseen

Stack (top at right): J | I | G | D | F

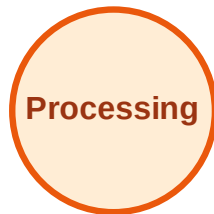
Visited: B, E



DFS Traversal

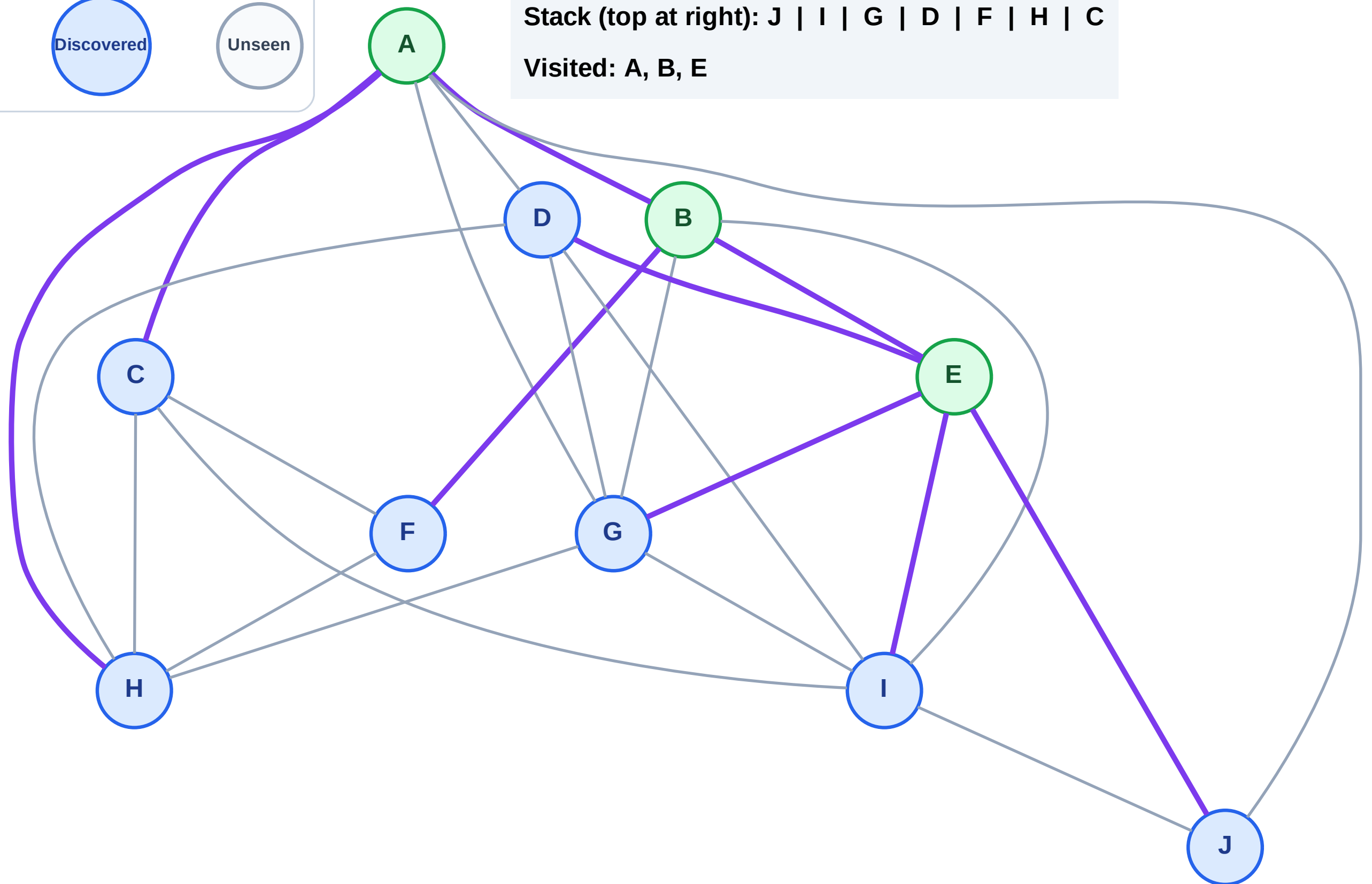
Step 7: Discover H, C from A

Legend



Stack (top at right): J | I | G | D | F | H | C

Visited: A, B, E



DFS Traversal

Step 8: Process node C

Legend

Complete

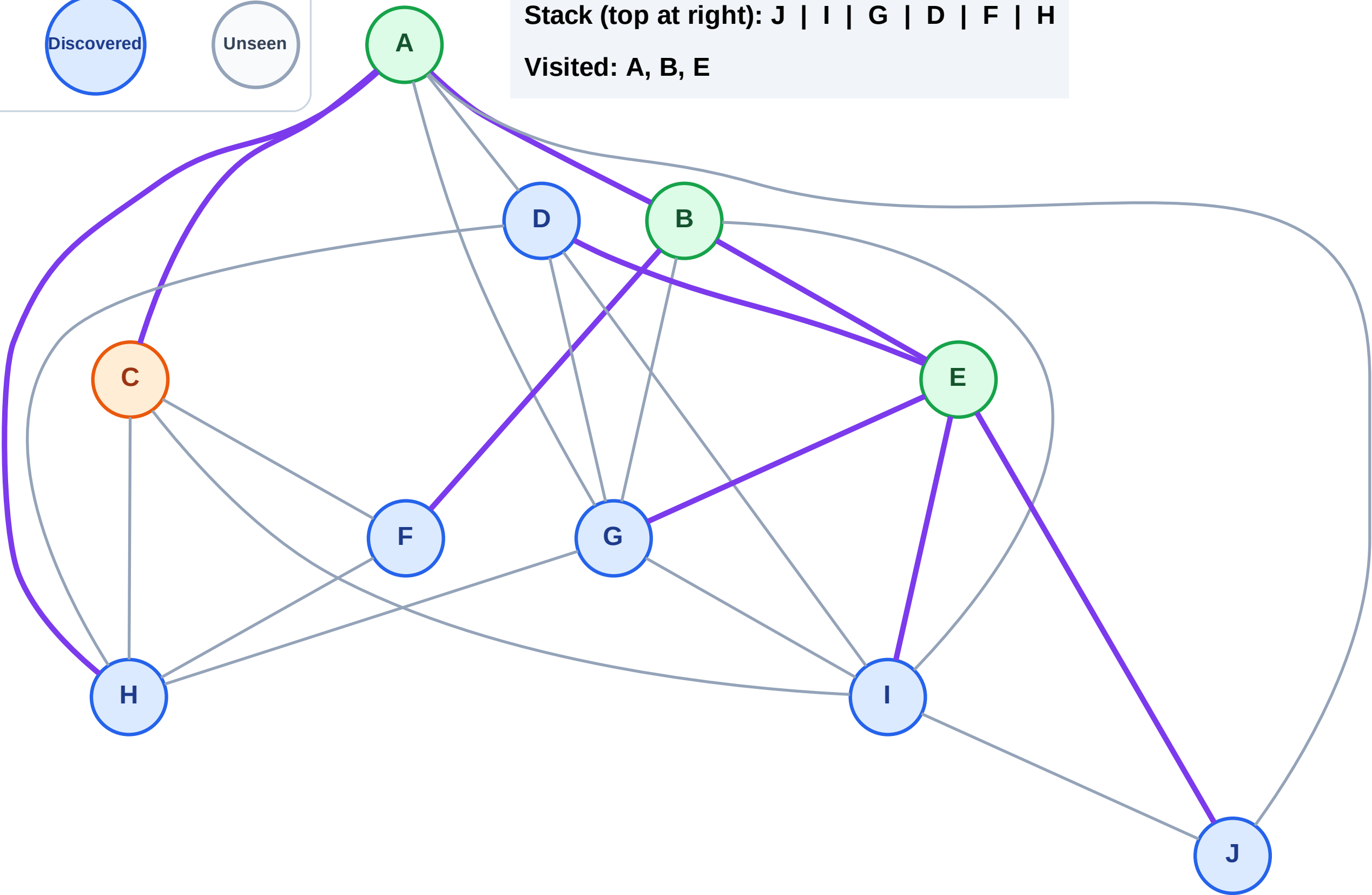
Processing

Discovered

Unseen

Stack (top at right): J | I | G | D | F | H

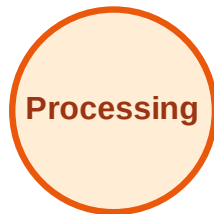
Visited: A, B, E



DFS Traversal

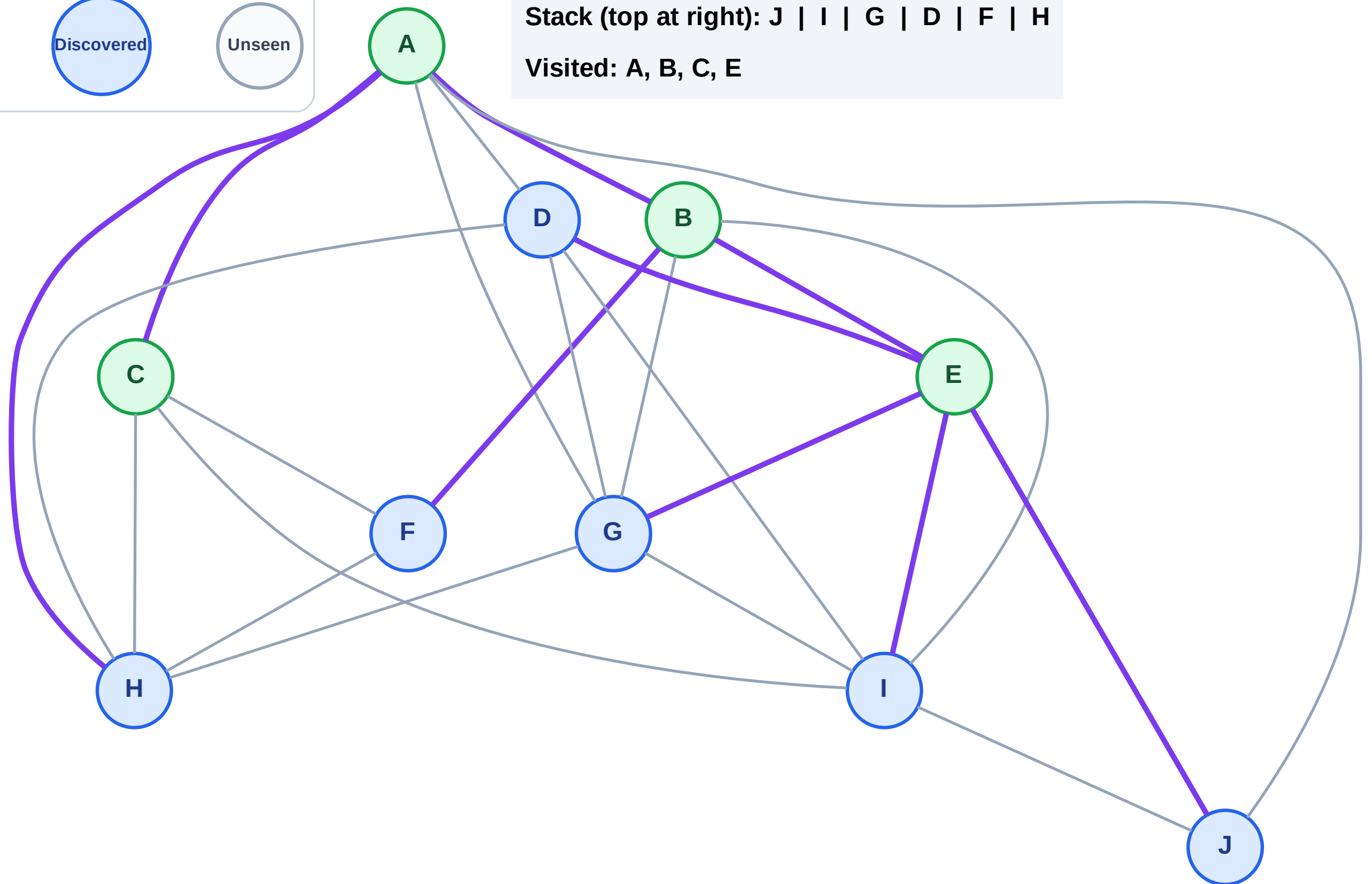
Step 9: No new neighbors from C

Legend



Stack (top at right): J | I | G | D | F | H

Visited: A, B, C, E



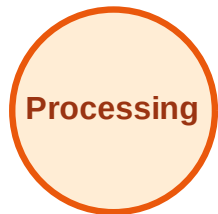
DFS Traversal

Step 10: Process node H

Legend



Complete



Processing



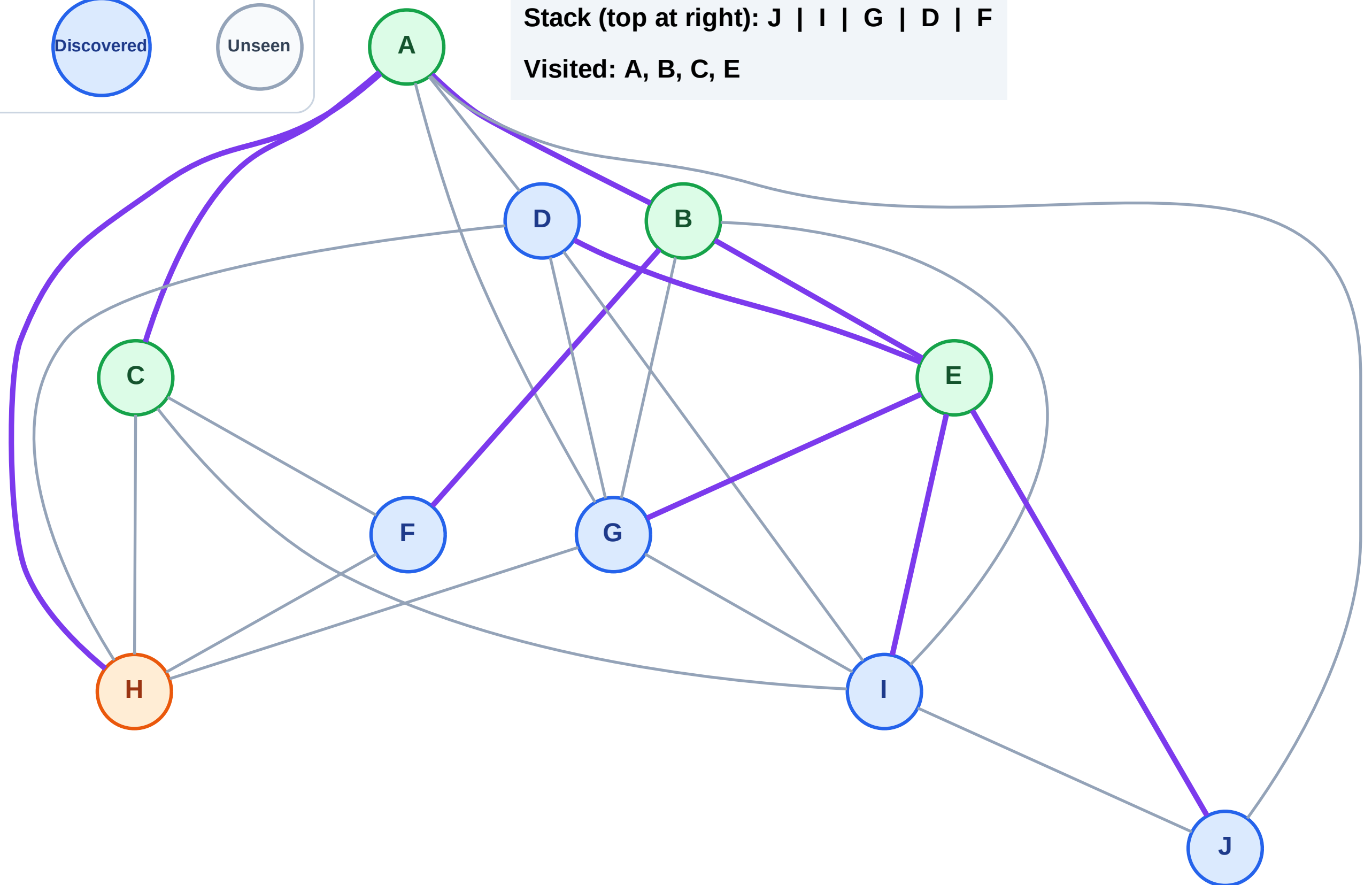
Discovered



Unseen

Stack (top at right): J | I | G | D | F

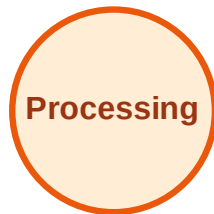
Visited: A, B, C, E



DFS Traversal

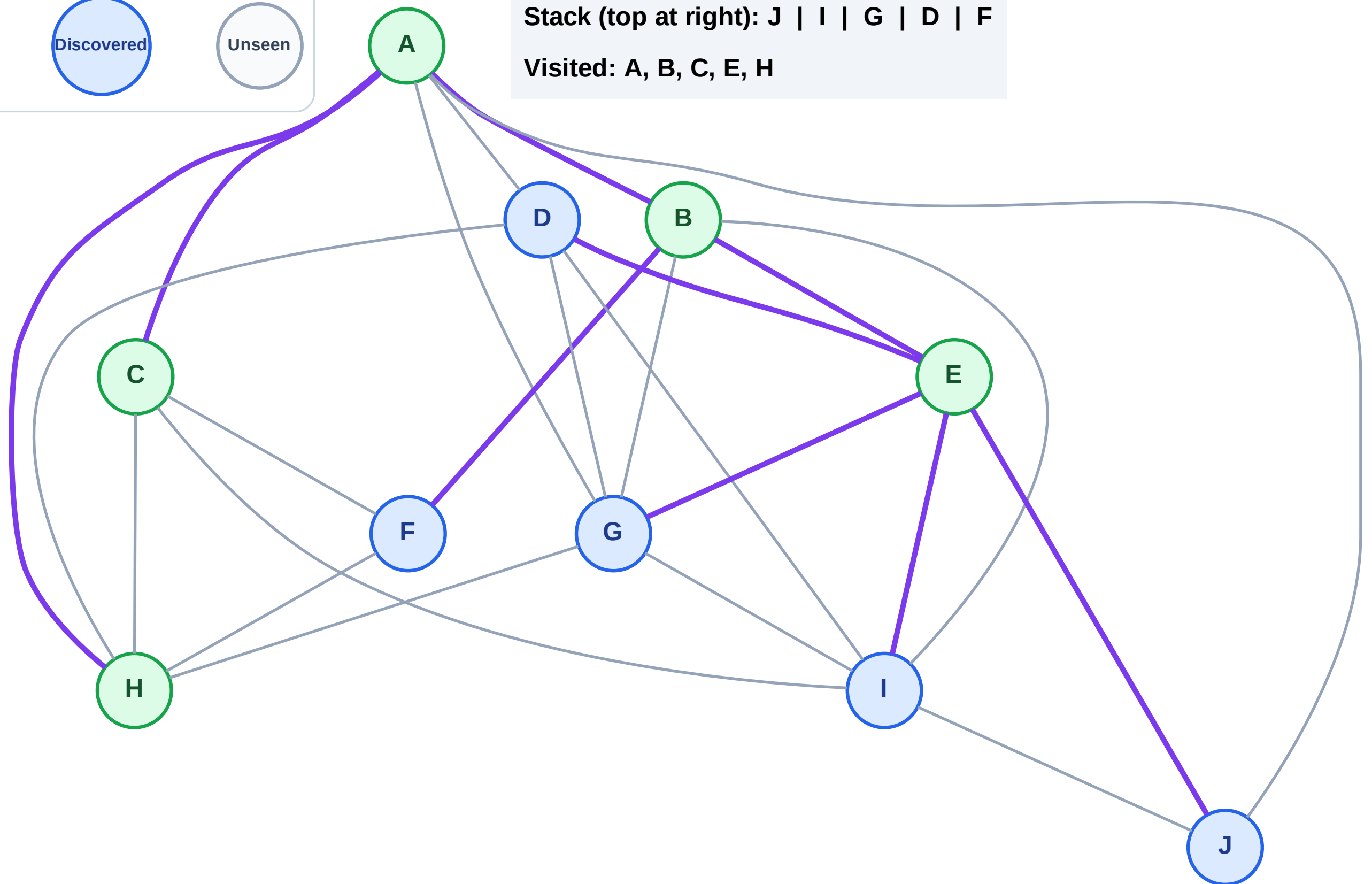
Step 11: No new neighbors from H

Legend



Stack (top at right): J | I | G | D | F

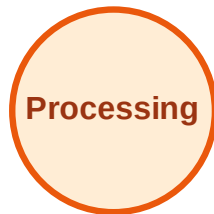
Visited: A, B, C, E, H



DFS Traversal

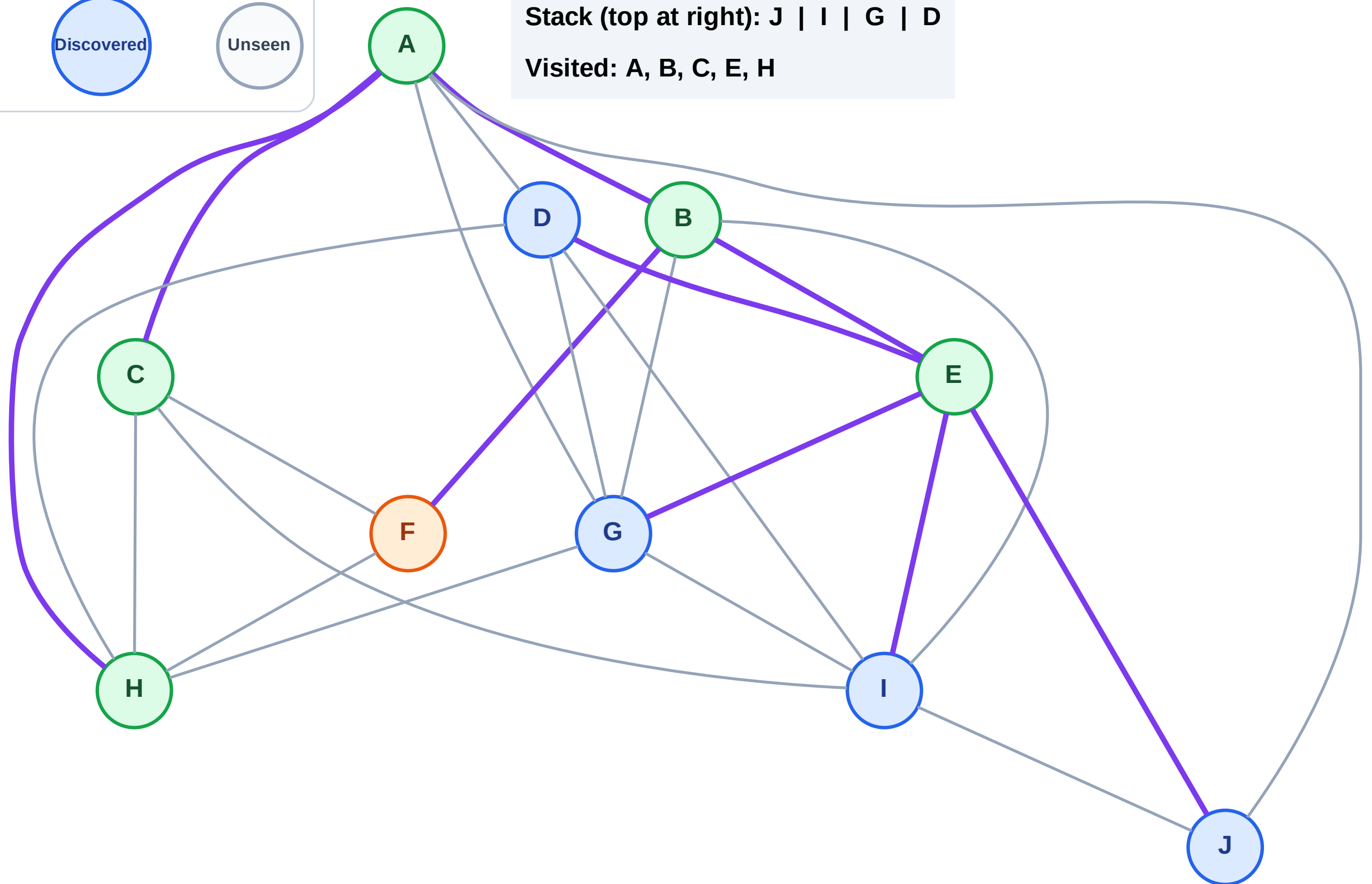
Step 12: Process node F

Legend



Stack (top at right): J | I | G | D

Visited: A, B, C, E, H



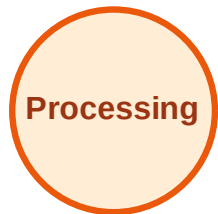
DFS Traversal

Step 13: No new neighbors from F

Legend



Complete



Processing



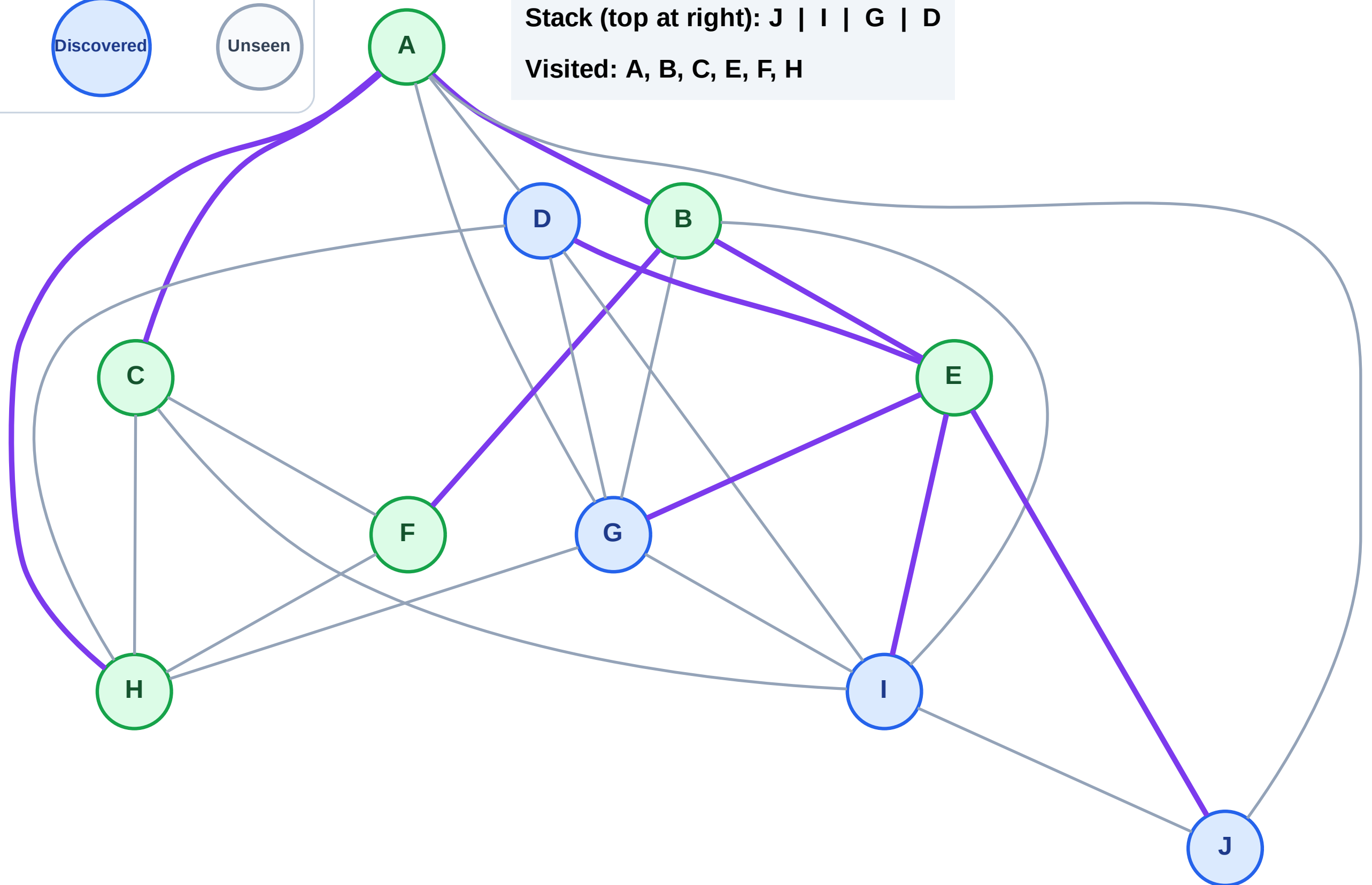
Discovered



Unseen

Stack (top at right): J | I | G | D

Visited: A, B, C, E, F, H



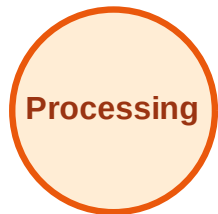
DFS Traversal

Step 14: Process node D

Legend



Complete



Processing



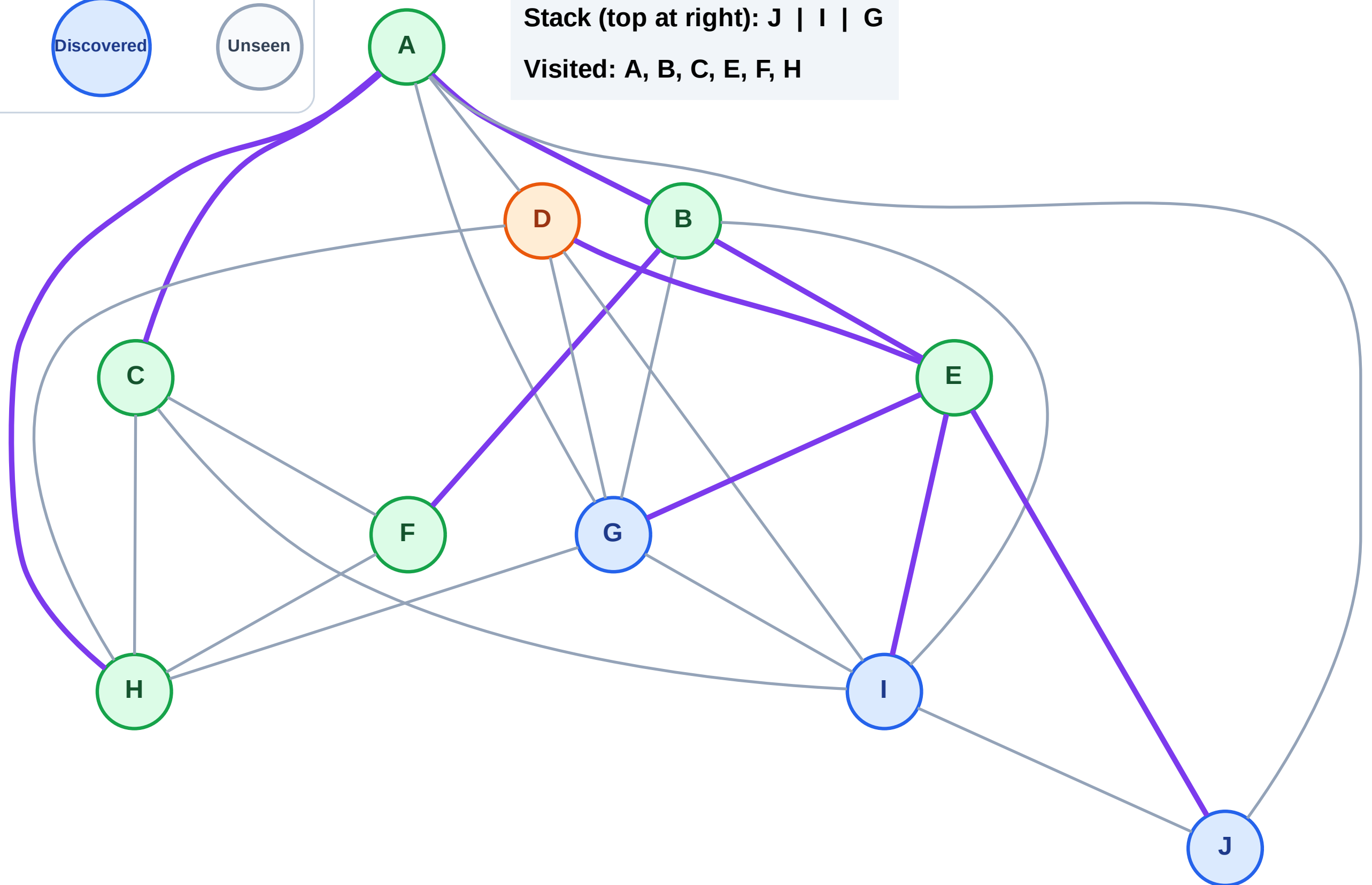
Discovered



Unseen

Stack (top at right): J | I | G

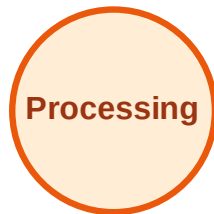
Visited: A, B, C, E, F, H



DFS Traversal

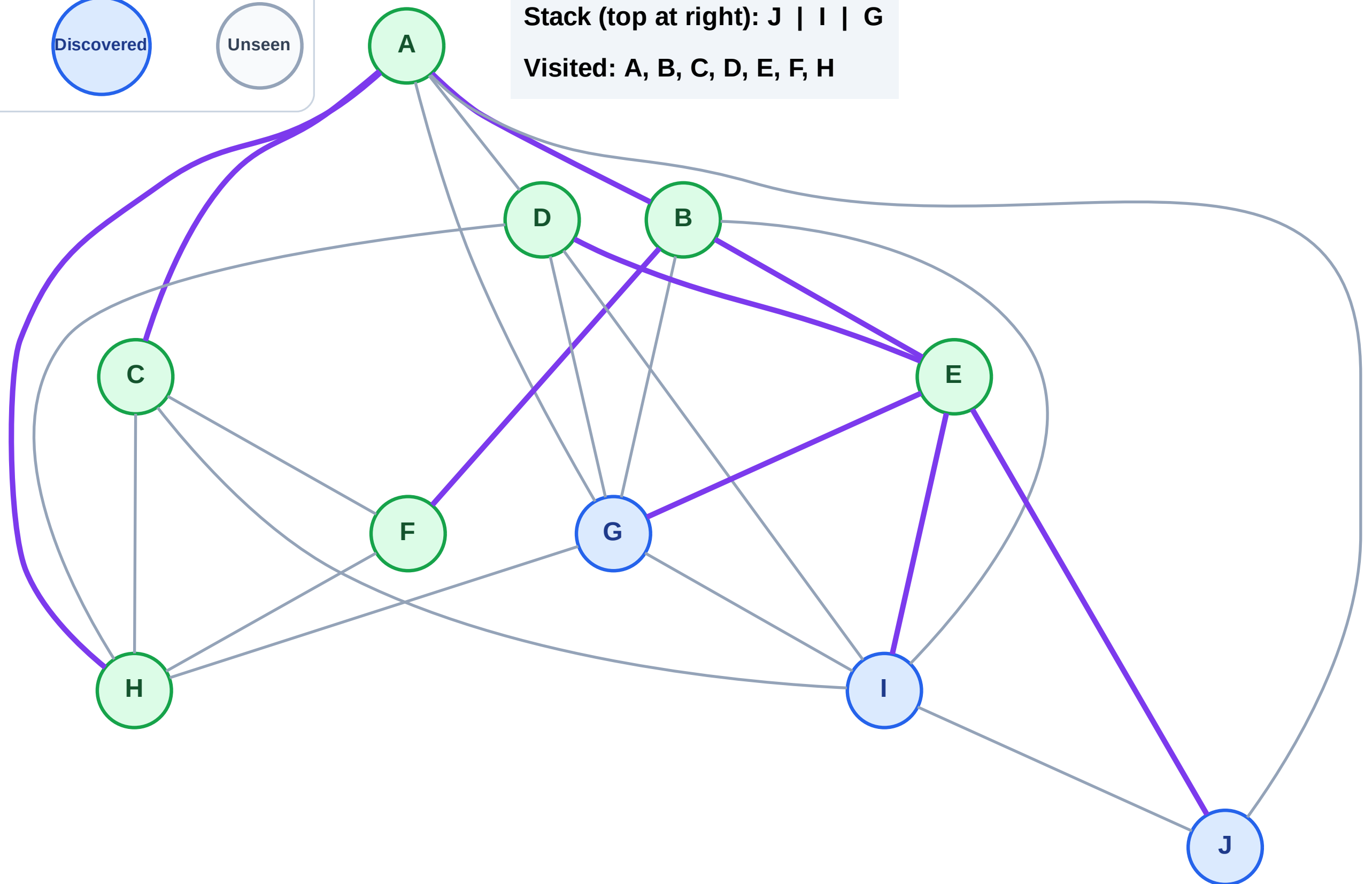
Step 15: No new neighbors from D

Legend



Stack (top at right): J | I | G

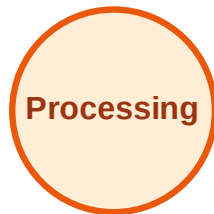
Visited: A, B, C, D, E, F, H



DFS Traversal

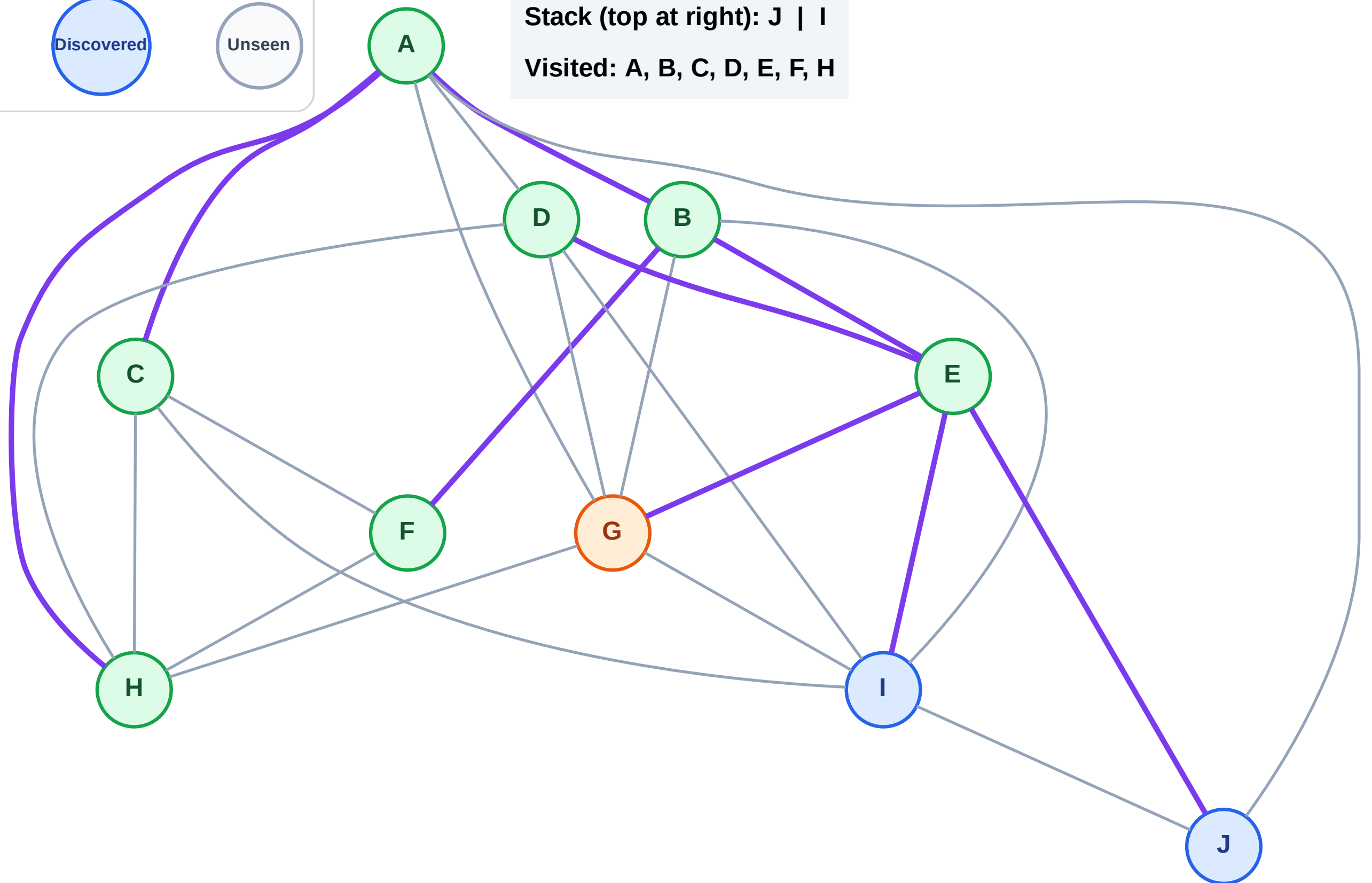
Step 16: Process node G

Legend



Stack (top at right): J | I

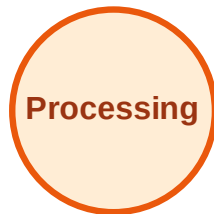
Visited: A, B, C, D, E, F, H



DFS Traversal

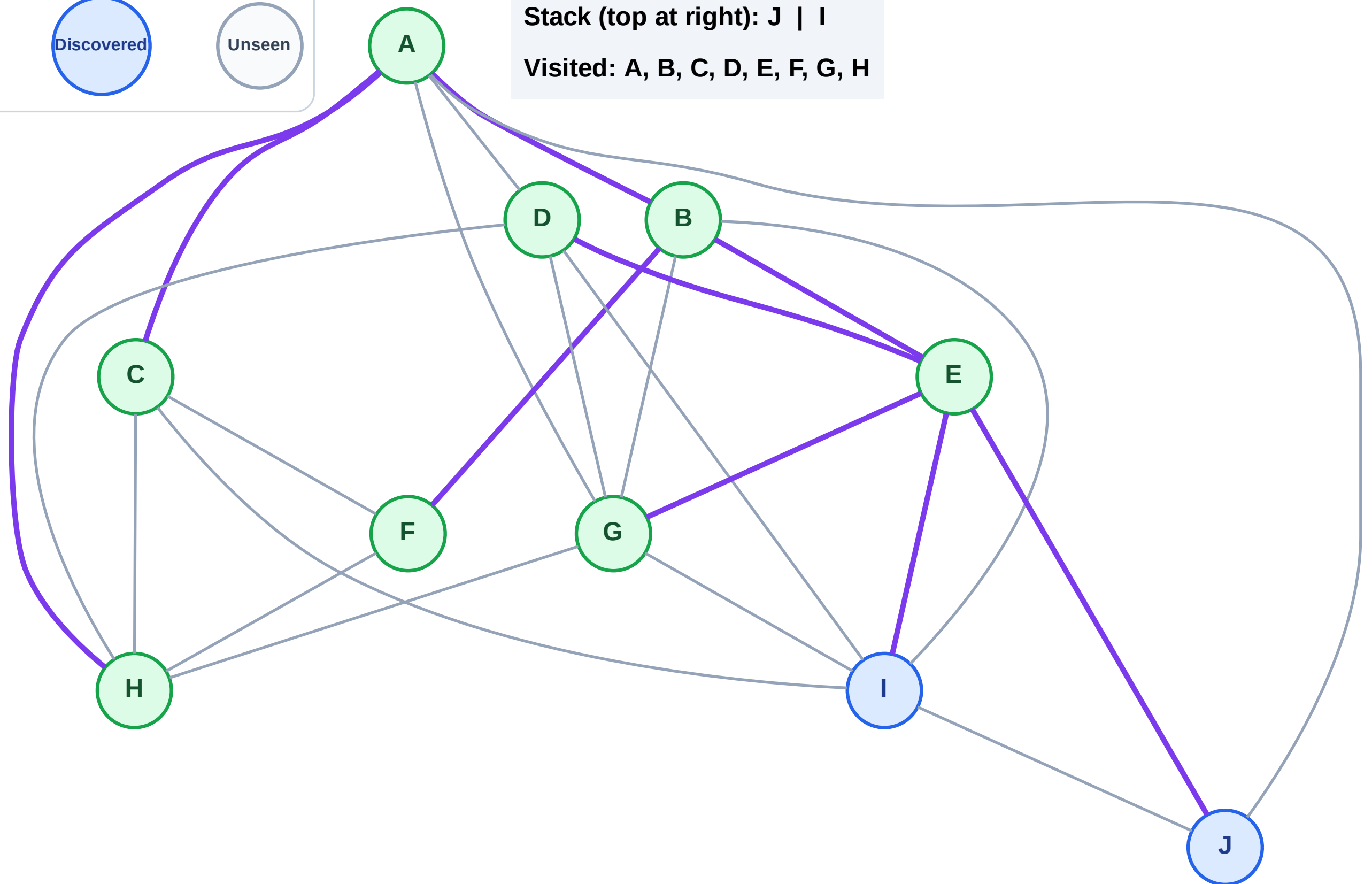
Step 17: No new neighbors from G

Legend



Stack (top at right): J | I

Visited: A, B, C, D, E, F, G, H



DFS Traversal

Step 18: Process node I

Legend

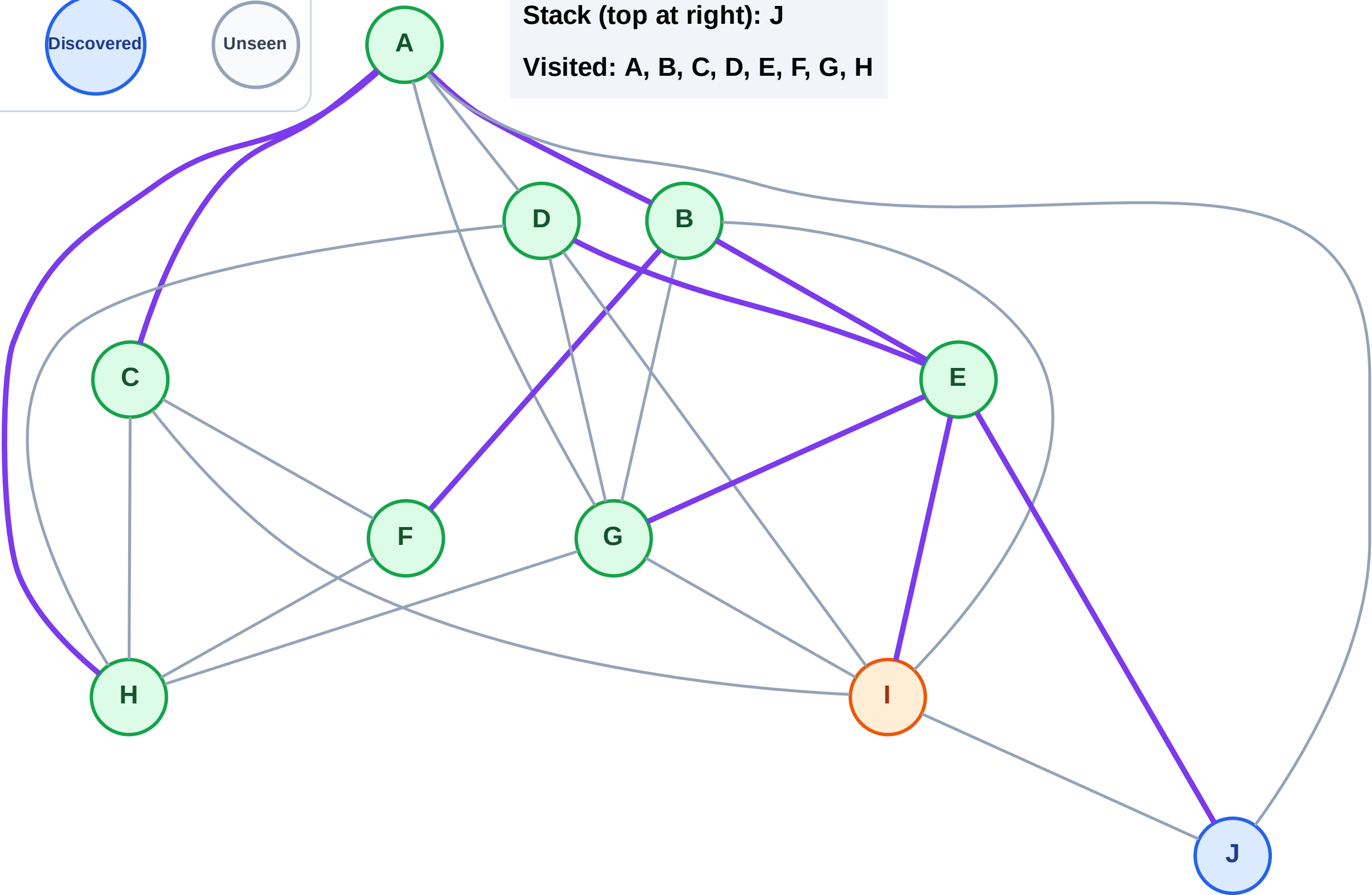
Complete

Processing

Discovered

Unseen

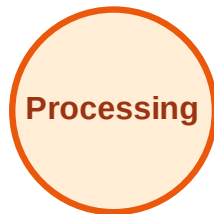
Stack (top at right): J
Visited: A, B, C, D, E, F, G, H



DFS Traversal

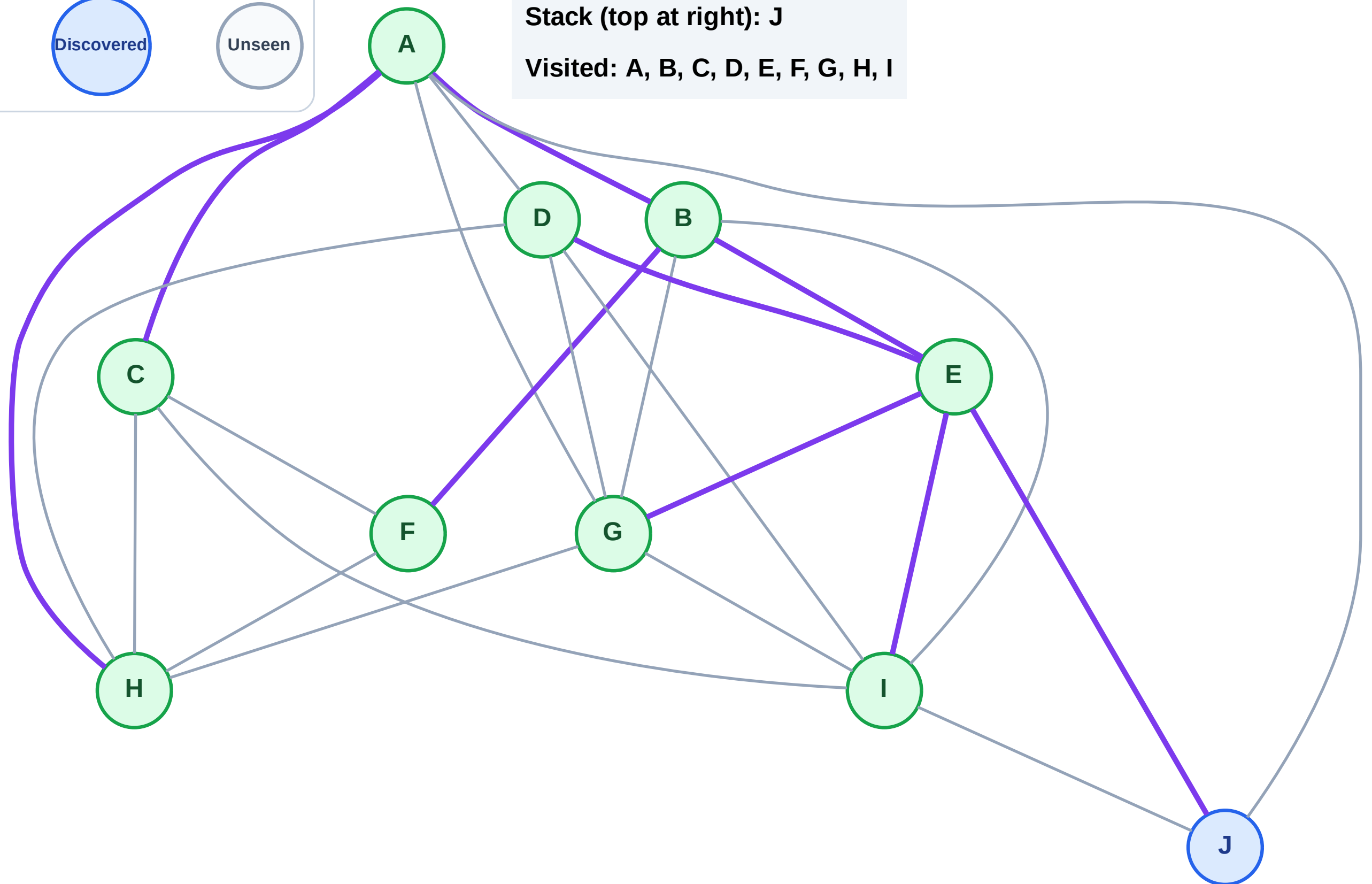
Step 19: No new neighbors from I

Legend



Stack (top at right): J

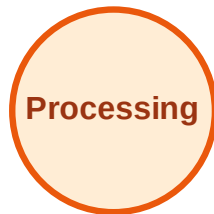
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

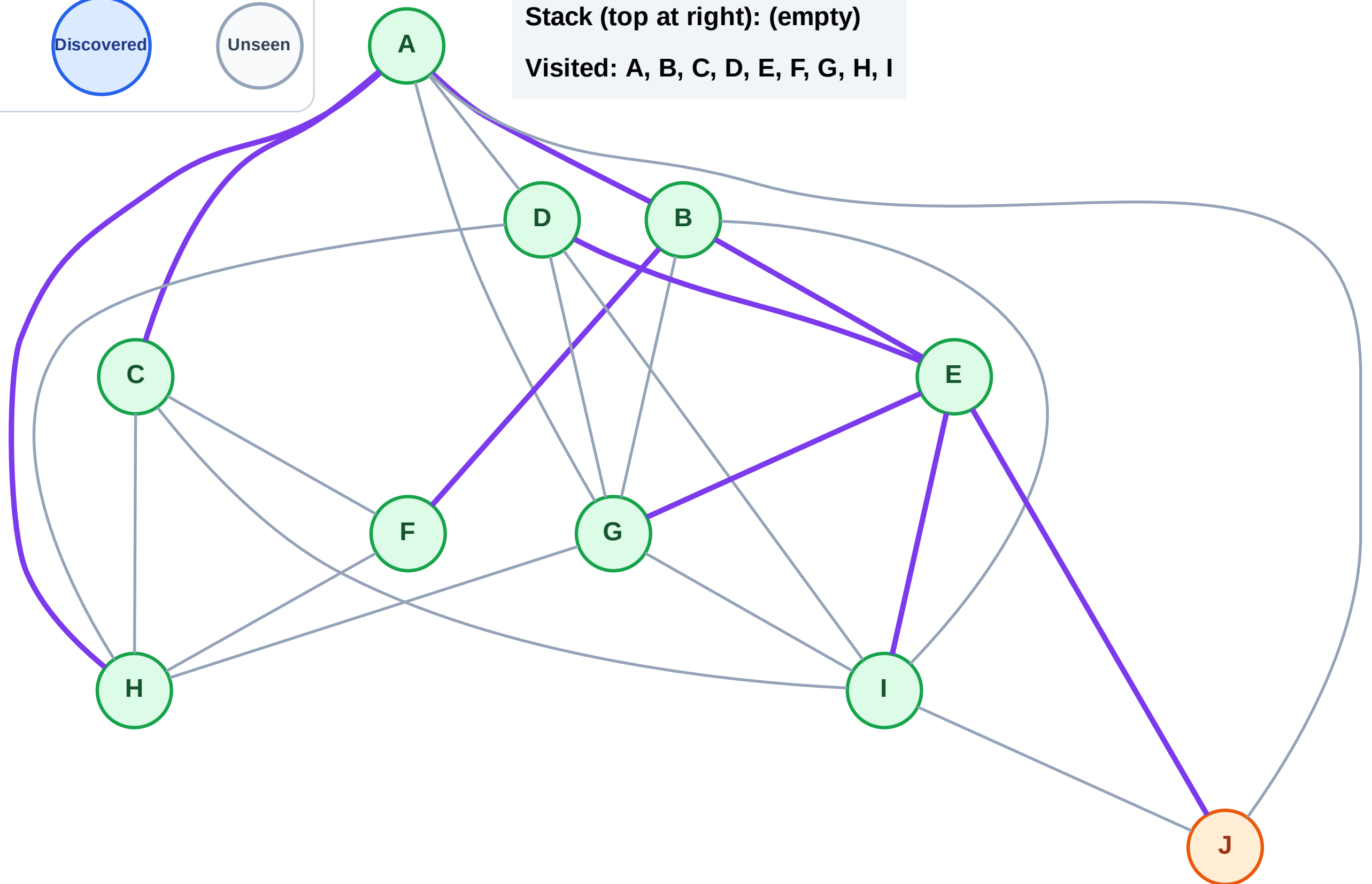
Step 20: Process node J

Legend



Stack (top at right): (empty)

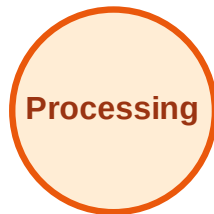
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 21: No new neighbors from J

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

